

Tape Spring Hinge

Lab Report

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May, 2026

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Abstract

This report investigates the geometric, dynamic, deployment, and bending behavior of a carbon-steel tape spring hinge using analytical theory, digital image correlation (DIC), MATLAB post-processing, Instron testing, and Abaqus simulation. The analytical model predicted that the localized bending radius of the tape spring should be approximately equal to its original transverse radius of curvature. DIC point-cloud cylinder fitting supported this prediction for the bent configurations, giving radii of 16.5077 mm for equal-sense bending and 15.8788 mm for opposite-sense bending, compared with the nominal 16.25 mm reference radius. The flat/deployed configuration produced a smaller fitted radius of 13.4037 mm, likely due to nonuniformity in the actual tape spring shape.

The vibration response was analyzed using both the DIC software FFT tool and a separate MATLAB FFT algorithm. The MATLAB FFT identified dominant peaks at 50.331 Hz, 111.70 Hz, and 204.415 Hz, which agreed reasonably with the interpolated DIC software FFT peaks near 50 Hz, 115 Hz, and 205 Hz. These were compared with Abaqus frequency predictions of 62.746 Hz, 117.20 Hz, and 266.20 Hz. The second mode showed the closest agreement, while discrepancies in the first and third modes were attributed to fixture compliance, stand dynamics, paint and speckle effects, manual tapping, and simplified Abaqus boundary conditions.

Deployment was evaluated qualitatively using selected high-speed image frames, showing rapid elastic recovery, overshoot, post-deployment oscillation, and damping to a steady deployed state. The 3-point bending experiment showed nonlinear moment-angle behavior, including a sharp positive peak followed by a lower steady-state plateau. Overall, the tape spring followed the expected curvature, vibration, deployment, and bending trends, while differences between theory, experiment, and simulation were mainly caused by geometric nonlinearity, simplified modeling assumptions, fixture interaction, and experimental uncertainty.

1. Introduction

Tape springs are thin, curved metallic strips that can undergo large elastic deformation while still recovering their original deployed shape. Unlike conventional straight beams, their mechanical response is strongly influenced by their initial transverse curvature. This curvature allows the structure to store elastic strain energy during bending and then release that energy during deployment. Because of this behavior, tape springs are commonly used as hinge or deployment elements in lightweight deployable structures, especially in space applications where compact stowage, passive deployment, and low structural mass are important.

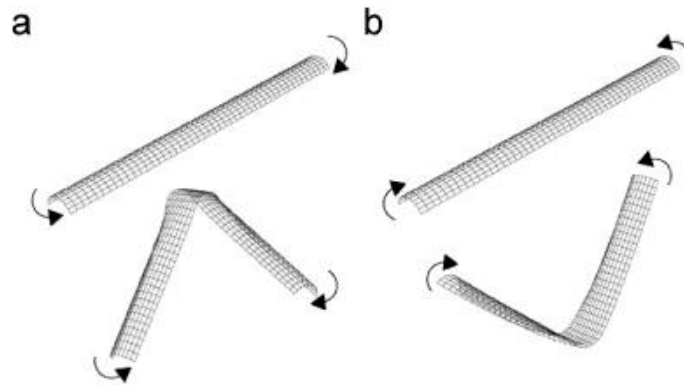


Figure 1. Tape spring hinge bending configurations: (a) equal-sense and (b) opposite-sense.

In this lab, the structure of interest is a tape spring hinge made from carbon steel. The tape spring has a thickness of 120 μm , a length of 8 inches, an arc length of 1 inch, and an initial radius of curvature of 16.25 mm. The material is modeled as carbon steel with an elastic modulus of 200 GPa and Poisson's ratio of 0.3. During deformation, the tape spring can be bent in both equal-sense and opposite-sense directions, Figure 1. Equal-sense bending refers to bending in the same sense as the original transverse curvature, while opposite-sense bending refers to bending against the original curvature. These two cases produce different curvature changes, moment-angle behavior, and deformation characteristics, making them important to study separately.

The purpose of this report is to evaluate the tape spring hinge using a combination of theoretical analysis, digital image correlation, mechanical testing, and finite element simulation. First, DIC measurements are used to capture the three-dimensional geometry of the tape spring in the flat, equal-sense bending, and opposite-sense bending configurations. The measured point clouds are fitted to cylindrical surfaces so that the experimental radius of curvature can be compared with the theoretical prediction. This provides a direct validation of the curvature model and also shows how closely the real tape spring behavior follows the ideal geometric assumptions.

The dynamic response of the tape spring is then studied using high-speed DIC. During deployment, selected images at known timestamps are used to describe the qualitative motion of the structure as it releases from a deflected configuration and vibrates after deployment. A separate vibration dataset is analyzed using FFT methods in MATLAB, with displacement histories taken from selected DIC points away from the noisy edge regions. The resulting frequency peaks are compared with the FFT output from the DIC software and with an Abaqus frequency simulation. This comparison is used to identify the dominant bending and twisting modes of the deployed tape spring.

Finally, the quasi-static bending response is studied through 3-point bending tests in both equal-sense and opposite-sense bending. The measured Instron force and crosshead displacement are converted into bending moment and bending angle, allowing the experimental moment-angle curves to be compared with analytical theory and Abaqus bending simulations. Through these combined measurements and models, this report aims to determine whether the tape spring follows the expected curvature, vibration, and moment-angle behavior, while also identifying the main sources of discrepancy between theory, experiment, and simulation.

2. Theory and Analytical Model

This portion of the report is used to establish the theoretical references for the geometric, dynamic, and bending behavior of the tape spring. Referring to the instructions by Yao, this lab requires four main comparisons and analysis. The DIC-measured curvatures are compared with the expected tape spring curvature, the vibration frequencies extracted from DIC displacement data are compared with numerical results from Abaqus, deployment images are analyzed qualitatively using the physical stages of release until damping, and finally the 3-point bending moment-angle response from analytical calculation, numerical simulation, and experiment are compared.

2.1. Radius of Curvature

The tape spring is initially a thin curved strip with transverse radius of curvature R , length L , thickness t , and subtended angle α . When the tape spring is bent into a localized hinge shape, the radius of curvature of the bent region is denoted as r . The objective of this part of the theory is to determine the expected value of r for equal-sense and opposite-sense bending so that it can later be compared with the radius obtained from DIC cylinder fitting.

For a thin isotropic tape spring, the plate bending stiffness is

$$D = \frac{Et^3}{12(1 - \nu^2)}$$

where E is the elastic modulus, t is the tape thickness, and ν is Poisson's ratio. During bending, the tape spring experiences a longitudinal curvature associated with the bending radius r , while the transverse curvature is associated with the original radius R . The curvature changes can be represented as

$$\Delta k_x = \pm \frac{1}{r}, \quad \Delta k_\theta = -\frac{1}{R}$$

Where the sign of Δk depends on whether the tape spring is bent in the equal-sense or opposite-sense direction. The transverse curvature change is negative because the tape spring locally flattens or reverses its original curved cross-section in the hinge region.

The bending strain energy per unit area can be written as

$$u = \frac{D}{2} [\Delta k_x^2 + \Delta k_\theta^2 + 2\nu \Delta k_x \Delta k_\theta]$$

Substituting the curvature terms gives

$$u = \frac{D}{2} \left[\frac{1}{r^2} + \frac{1}{R^2} \pm 2v \frac{1}{rR} \right]$$

The total strain energy is obtained by multiplying the energy density by the curved surface area of the bent region. If the bent region has length l and bending angle θ , then the arc length of the bend is $r\theta$, so the total strain energy becomes

$$u_{total} = l\theta r u = \frac{Dl\theta r}{2} \left[\frac{1}{r^2} + \frac{1}{R^2} \pm 2v \frac{1}{rR} \right]$$

$$u_{total} = \frac{Dl\theta}{2} \left[\frac{1}{r} + \frac{r}{R^2} \pm 2v \frac{1}{R} \right]$$

Equilibrium bending radius is found by minimizing the total strain energy with respect to r

$$\frac{d(u_{total})}{dr} = \frac{Dl\theta}{2} \left[\frac{d}{dr} \left(\frac{1}{r} \right) + \frac{d}{dr} \left(\frac{r}{R^2} \right) + \frac{d}{dr} \left(2v \frac{1}{R} \right) \right] = 0$$

$$\frac{d(u_{total})}{dr} = \frac{Dl\theta}{2} \left[-\frac{1}{r^2} + \frac{1}{R^2} + 0 \right] = 0$$

$$\frac{Dl\theta}{2} \frac{1}{R^2} = \frac{Dl\theta}{2} \frac{1}{r^2}$$

Therefore,

$$\frac{1}{R^2} = \frac{1}{r^2}$$

and the expected bending radius is

$$R = r$$

This result shows that, under the ideal thin-shell tape spring model, the bending radius of the localized hinge region is expected to be equal in magnitude to the original transverse radius of curvature. Therefore, the equal-sense and opposite-sense bending cases are expected to have the same theoretical radius magnitude, although the bending direction and moment sign are different. In the experiment, this value is compared with the radius extracted from the DIC point cloud by fitting the measured X_p , Y_p , and Z_p coordinates to a cylinder.

2.2. Fast Fourier Transform (FFT) and Vibration Frequency

The vibration behavior of the tape spring is analyzed using a manual vibration test. In this test, the tape spring is mounted in the deployed configuration and then excited by tapping the structure with a screwdriver. After the tap, the tape spring vibrates freely about its equilibrium position. The resulting displacement response contains the natural frequency content of the mounted tape spring system.

The purpose of the FFT analysis is to transform the measured displacement response from the time domain into the frequency domain. In the time domain, the DIC data describe how a selected point moves as a function of time. For a selected DIC point, the displacement history can be represented as

$$x[n] = x(t_n)$$

where $x[n]$ is the measured displacement at frame n , N is the total number of frames, and the sampling frequency is

$$f_s = \frac{1}{\Delta t}$$

where Δt is the time spacing between frames. Since DIC gives displacement components in the U , V , and W directions, the FFT can be applied to one or more of these displacement histories depending on which direction contains the clearest vibration response.

Before applying the FFT, the mean displacement is removed from the signal:

$$\tilde{x}[n] = x[n] - \bar{x}$$

This step removes the static offset of the mounted tape spring and makes the frequency plot focus on the oscillatory part of the response. The discrete Fourier transform of the mean-subtracted signal is then written as

$$X[k] = \sum_{n=0}^{N-1} \tilde{x}[n] e^{-i2\pi kn/N}$$

where $X[k]$ represents the contribution of the signal at frequency index k . Each frequency index corresponds to a physical frequency,

$$f_k = \frac{k f_s}{N}$$

The spacing between frequency bins is therefore

$$\Delta f = \frac{f_s}{N}$$

which means that a longer time signal provides finer frequency resolution. The highest frequency that can be resolved without aliasing is the Nyquist frequency,

$$f_{\text{Nyquist}} = \frac{f_s}{2}$$

The magnitude of the Fourier transform gives the frequency spectrum. For a single-sided amplitude spectrum, the amplitude can be written as

$$A[k] = \frac{2 |X[k]|}{N}$$

for the positive-frequency range, excluding the zero-frequency component. Peaks in this spectrum indicate frequencies that are strongly present in the measured vibration response. These peaks are interpreted as candidate natural frequencies of the tape spring.

Physically, this works because a vibrating structure can be described as a combination of modal responses. After the tape spring is tapped, several vibration modes may be excited at the same time. Each mode contributes an oscillatory component to the measured displacement signal. The FFT separates the total displacement history into its frequency components, allowing the dominant modal frequencies to be identified from the peaks in the spectrum.

This is the foundation of the FFT analysis that will be conducted in MATLAB, in which the results will be compared with two other frequency references: the internal FFT result from the DIC software and the natural frequencies predicted by Abaqus. The DIC software FFT provides a direct comparison to the MATLAB post-processing because both are based on the experimental displacement data. Abaqus provides a numerical modal prediction based on the modeled geometry, material properties, and boundary conditions. Agreement between these three approaches would suggest that the identified frequency peaks correspond to real structural modes of the mounted tape spring.

The main modes expected in the frequency response are bending and twisting modes. A bending mode is associated primarily with transverse flexural motion of the tape spring, while a twisting mode involves rotation of the curved cross-section about the length of the tape. Because the tape spring is curved and clamped, the modes may not appear as perfectly

separated pure bending or pure twisting. Instead, the measured vibration response may include coupled bending-twisting behavior. Additional smaller peaks may also appear due to higher modes, clamp vibration, imperfect tapping, fixture compliance, DIC tracking noise, or local point-cloud noise. For this reason, the most meaningful frequency peaks are the ones that appear clearly in the MATLAB FFT, are also visible in the DIC software FFT, and correspond reasonably with the Abaqus mode shapes.

2.3. Deployment Behavior

The qualitative deployment analysis is based on a separate pull-and-release test. In this test, the tape spring is mounted at one end, manually pulled back from its deployed configuration, and then released. This test is different from the manual vibration test used for FFT analysis. The vibration test is intended to identify natural frequencies after the mounted tape spring is tapped, while the deployment test is intended to observe how the structure physically recovers from a deflected state. Therefore, the deployment image sequence is interpreted qualitatively.

The deployment process can be understood as an energy conversion problem. Before release, the tape spring is held in a deformed configuration and stores elastic strain energy due to bending and local curvature change. When the external constraint is removed, this stored strain energy is converted into kinetic energy, causing the tape spring to accelerate back toward its natural deployed shape. Because the tape spring is thin, curved, and geometrically nonlinear, the recovery motion is not expected to behave like a simple rigid-body rotation. Instead, the structure may show localized bending, cross-section flattening, bending-twisting coupling, overshoot, residual oscillation, and damping.

For the qualitative discussion, the deployment process can be divided into several physical stages. The first stage is the pre-release or loaded stage. In this stage, the tape spring is still being held in the pulled-back position. The structure is not moving freely yet, but elastic strain energy has already been stored by the imposed deformation. The exact shape in this stage depends on the pulling direction, the amount of deflection, and the constraint applied by the mounted end.

The second stage is the release and rapid deployment stage. Once the tape spring is released, the stored elastic energy begins converting into kinetic energy. The tape spring accelerates toward its deployed configuration, and the motion is expected to occur quickly because of the low mass and elastic recovery of the thin curved strip. During this stage, the motion may

involve both bending and twisting, especially if the release is not perfectly symmetric or if the clamp introduces slight off-axis constraints.

The third stage is the first deployed or overshoot stage. At this point, the tape spring has returned close to its deployed shape. However, because the structure still has inertia, it may not stop exactly at the equilibrium configuration. Instead, it can pass through the deployed position and overshoot before reversing direction. This overshoot is important because it shows that the deployment is dynamic rather than purely quasi-static.

The fourth stage is the post-deployment oscillation stage. After the main deployment motion is complete, the tape spring continues to vibrate about its deployed equilibrium shape. This visible oscillation is caused by the remaining kinetic energy in the structure after release. Although this stage may appear similar to the tapped vibration response used for FFT analysis, the excitation source is different. In the deployment test, the oscillation is produced by pull-and-release recovery, while in the vibration test it is produced by a deliberate tap on the mounted tape spring.

The final stage is the damping and steady-state stage. Over time, the oscillation amplitude decreases as mechanical energy is dissipated through material damping, clamp friction, small fixture motion, air resistance, and other non-ideal effects. Eventually, the tape spring settles into its steady deployed configuration. In the report, the high-speed image sequence can therefore be discussed using these stages: before release, during rapid deployment, near first full deployment, during visible oscillation, and after the motion has mostly damped out.

This phase-based interpretation provides a structured way to discuss the deployment images. Instead of only describing what each frame looks like, the image sequence can be used to explain how the tape spring transitions from stored elastic deformation, to rapid recovery, to overshoot and oscillation, and finally to a damped steady state.

2.4. Moment-Angle Behavior

The final theory is the expected moment-angle behavior of the tape spring during equal-sense and opposite-sense bending. The analytical steady-state moment can be derived from the isotropic plate bending relation,

$$\begin{bmatrix} M_x \\ M_\theta \end{bmatrix} = \frac{Et^3}{12(1-\nu^2)} \begin{bmatrix} 1 & \nu \\ \nu & 1 \end{bmatrix} \begin{bmatrix} \Delta k_x \\ \Delta k_\theta \end{bmatrix}$$

where M_x is the longitudinal bending moment per unit width, M_θ is the transverse bending moment per unit width, E is the elastic modulus, t is the tape thickness, and ν is the Poisson's ratio. Using the equilibrium result that the bending radius is equal in magnitude to the original transverse radius, $r = R$, the curvature-change vector can be written as

$$\begin{bmatrix} \Delta k_x \\ \Delta k_\theta \end{bmatrix} = \begin{bmatrix} \pm \frac{1}{R} \\ -\frac{1}{R} \end{bmatrix}$$

where the positive or negative sign depends on whether the tape spring is bent in the equal-sense or opposite-sense direction. The longitudinal bending moment per unit width is therefore

$$\begin{bmatrix} M_x \\ M_\theta \end{bmatrix} = \frac{Et^3}{12(1-\nu^2)} \begin{bmatrix} 1 & \nu \\ \nu & 1 \end{bmatrix} \begin{bmatrix} \pm \frac{1}{R} \\ -\frac{1}{R} \end{bmatrix} = \frac{Et^3}{12(1-\nu^2)} \begin{bmatrix} \pm 1 - \nu \\ \pm \nu - 1 \end{bmatrix}$$

$$M_x = \frac{Et^3}{12(1-\nu^2)} \left[\frac{\pm 1 - \nu}{R} \right]$$

The total bending moment is found by multiplying the moment per unit width by the arc width of the tape spring. Since the arc width is αR , where α is the subtended angle in radians,

$$M_{\text{total}} = M_x \alpha R$$

Substituting the expression for M_x gives

$$M_{\text{total}} = \frac{Et^3 \alpha}{12(1-\nu^2)} (\pm 1 - \nu)$$

Thus, the steady-state moments are

$$M_+ = \frac{Et^3 \alpha}{12(1-\nu^2)} (1 + \nu)$$

and

$$M_- = -\frac{Et^3 \alpha}{12(1-\nu^2)} (1 - \nu)$$

The negative sign indicates that the moment acts in the opposite direction. Therefore, when comparing moment magnitudes, the absolute value of M_- is used.

Using the analytical calculation values and α from DIC,

$$E = 200 \text{ GPa}$$

$$\nu = 0.3$$

$$t = 120 \text{ } \mu\text{m}$$

$$\alpha = 87.14^\circ = 1.521 \text{ rad}$$

the predicted steady-state moments are

$$M_+ = +6.26 \times 10^{-2} \text{ N} \cdot \text{m}$$

and

$$M_- = -3.37 \times 10^{-2} \text{ N} \cdot \text{m}$$

These values show that the opposite-sense bending case is expected to require a larger steady-state moment than the equal-sense bending case. This occurs because the curvature terms contribute differently depending on the bending direction. In equal-sense bending, the curvature interaction reduces the effective moment contribution, while in opposite-sense bending the curvature changes produce a larger moment magnitude. These analytical steady-state moments are used later as reference values for the experimental 3-point bending curves and the Abaqus bending simulation. The material properties, tape thickness, and subtended angle used in this calculation are taken from the analytical calculation document.

In the 3-point bending test, the Instron machine records the applied force and crosshead displacement. These measured quantities must be converted into bending moment and bending angle before the experimental response can be compared with the analytical prediction and Abaqus bending simulation. The lab module specifies that this test should be performed for both equal-sense and opposite-sense bending, using a lower anvil span of 120 mm, a travel distance of 25 mm, and a loading rate of 10 mm/min.

For a simply supported 3-point bending configuration, the maximum bending moment occurs at the center loading point. If the applied force is F and the lower support span is S , the center bending moment is

$$M = \frac{FS}{4}, \quad S = 120 \text{ mm}$$

The bending angle can be estimated from the crosshead displacement. If the deformed tape spring is approximated as two straight segments meeting at the center loading point, then the bending angle is

$$\theta = 2 \tan^{-1} \left(\frac{2\delta}{S} \right)$$

where δ is the crosshead displacement. This approximation converts the force-displacement data into a moment-angle curve. Since the displacement is taken from the Instron crosshead, the calculated angle should be interpreted as an effective bending angle rather than a perfectly local measurement of the tape spring deformation. Small errors may arise from fixture compliance, local contact effects, or slipping at the supports.

The expected moment-angle response of a tape spring is nonlinear. At small bending angles, the moment increases as the tape spring begins to deform from its original curved geometry. As bending continues, the tape spring can form a localized hinge region where the cross-section partially flattens or changes curvature. Once this hinge-like region develops, additional rotation can occur with an approximately constant moment. This produces a steady-state moment region in the moment-angle plot. The equal-sense and opposite-sense bending cases are expected to have different steady-state moment magnitudes because the imposed longitudinal curvature interacts differently with the original transverse curvature of the tape spring.

3. Methods

The methods used in this report were organized to evaluate the tape spring through its geometric, dynamic, deployment, and bending behavior. The first part of the method focused on DIC-based radius measurement, where the three-dimensional surface geometry of the tape spring was captured in the flat/deployed, equal-sense bending, and opposite-sense bending configurations, then processed in MATLAB using cylinder fitting to obtain the experimental radius of curvature. The second part focused on vibration testing, where a separate tapped-response test was analyzed using MATLAB FFT processing and compared with the DIC software's internal FFT results. An Abaqus frequency model was also created to provide numerical predictions of the first vibration modes for comparison with the experimental frequencies.

In addition to the radius and vibration measurements, the deployment behavior was studied qualitatively using a separate pull-and-release test. This test was used to observe the physical sequence of deployment rather than to extract frequency data. Finally, the quasi-static bending behavior of the tape spring was measured using 3-point bending tests in both equal-sense and opposite-sense directions. The recorded force and displacement data were converted into moment-angle curves, which were then compared with analytical steady-state moment predictions and Abaqus bending simulations.

3.1. DIC Curvature Measurement

The radius of curvature of the tape spring was measured experimentally using 3D digital image correlation. This measurement was performed for three configurations: the original flat/deployed tape spring, the equal-sense bending configuration, and the opposite-sense bending configuration. In all three cases, the DIC system was used to reconstruct the three-dimensional surface geometry of the tape spring, and MATLAB was then used to fit the reconstructed geometry to a cylinder in order to obtain the experimental radius of curvature.



Figure 2. Tape spring with speckle pattern.

Before testing, a speckle pattern was manually applied to the outer surface of the tape spring using a Sharpie marker, Figure 2. This produced the random high-contrast surface pattern required for the DIC software to track the tape spring surface. The tape spring was then mounted using 3D-printed fixtures and clamped to a stand in front of the stereo DIC camera system. To improve the quality of the reconstructed point cloud, the camera focus and lighting were adjusted before each capture, and the DIC system was calibrated using a reference plate.

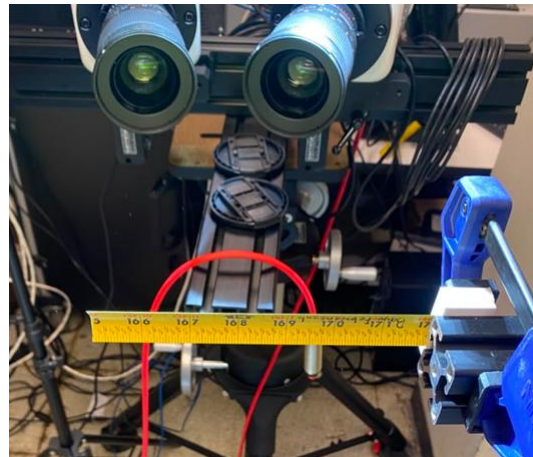


Figure 3. Flat tape spring DIC setup.

Three different 3D-printed mounts were used for the three measured configurations. For the flat/deployed configuration, the tape spring was placed in the flat mount so that the DIC cameras could capture the outer curved surface in its original state, Figure 3. For the equal-sense and opposite-sense configurations, separate mounts were used to hold the tape spring in the desired bending direction. In both bending cases, the tape spring was manually bent from its flat/deployed stable state into the appropriate fixture before the DIC capture was taken, Figure 4. This allowed the DIC system to measure the curved geometry while the tape spring was held in the equal-sense or opposite-sense bending configuration.

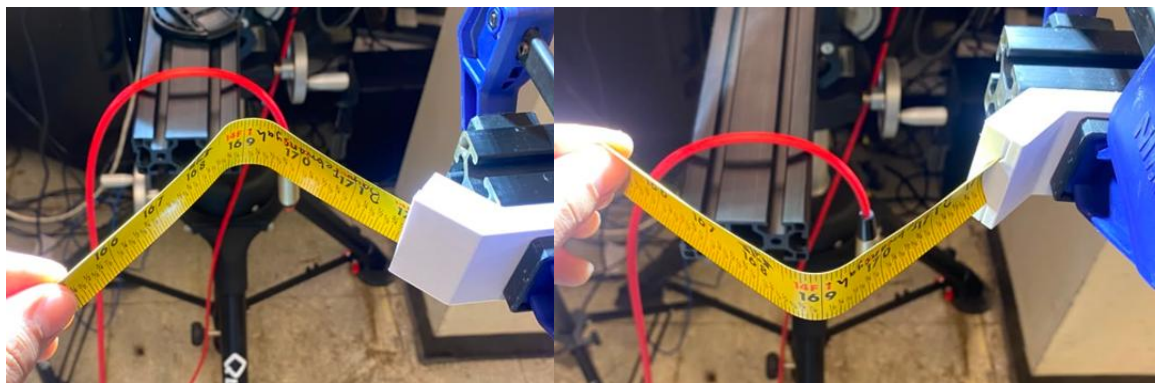


Figure 4. Equal-sense (left) and opposite-sense DIC setup.

After each DIC capture, the reconstructed geometry was exported as a data file containing the three-dimensional coordinates of the measured surface points. The coordinate columns used for the radius measurement were the final surface-position coordinates, corresponding to X_p , Y_p , and Z_p . These coordinates define the measured 3D point cloud of the tape spring surface.

The MATLAB processing was performed separately for the flat, equal-sense, and opposite-sense datasets. In each script, the coordinate data were imported, and only the last three coordinate columns were kept for the fitting process. Rows containing invalid or non-finite values were removed before creating the point cloud. This ensured that missing or corrupted DIC points did not affect the cylinder fitting result. The filtered coordinate data were then converted into a MATLAB pointCloud object.

A reference cylinder-axis direction was estimated using principal component analysis. The first principal component of the measured point cloud was used as the initial reference vector for the cylinder fit. This helped guide the `pcfitcylinder` function toward the dominant axis direction of the tape spring surface. The cylinder was then fitted using a maximum fitting distance of 0.01, and the MATLAB output provided the fitted cylinder radius, center location, cylinder orientation, number of inlier points, total number of measured points, and mean fitting error. The fitted radius was taken as the experimental radius of curvature for that configuration, while the mean fitting error was used as an indicator of fit quality.

In addition to calculating the radius, the MATLAB scripts were also used to visualize the fitting result. The measured DIC points were projected into a local two-dimensional PCA plane and triangulated using Delaunay triangulation to create a shell-like representation of the measured tape spring surface. Overly large triangles were removed so that the plotted shell more closely represented the actual measured geometry. A transparent fitted cylinder was then plotted over the measured shell, together with the fitted cylinder axis. This visualization was used to check whether the cylinder model reasonably matched the measured DIC surface before using the fitted radius in the final comparison.

This process was repeated for the flat/deployed, equal-sense, and opposite-sense configurations. The resulting radii were then compared with the analytical prediction from Section 2.1, where the ideal bending radius of the tape spring is expected to be equal in magnitude to the original transverse radius of curvature.

3.2. Vibration Testing

The vibration behavior of the tape spring was measured using a separate dynamic test from the deployment experiment. In this test, the tape spring was mounted in its flat/deployed configuration relative to the DIC cameras so that the cameras captured the vibration response from the outer part of the tape spring, Figure 5. The same general DIC setup procedure was used as in the radius measurement, including camera focus adjustment, lighting optimization, and calibration using a reference plate. However, instead of capturing a single static image, the DIC system recorded a high-speed image sequence at a sampling frequency of 2000 Hz.

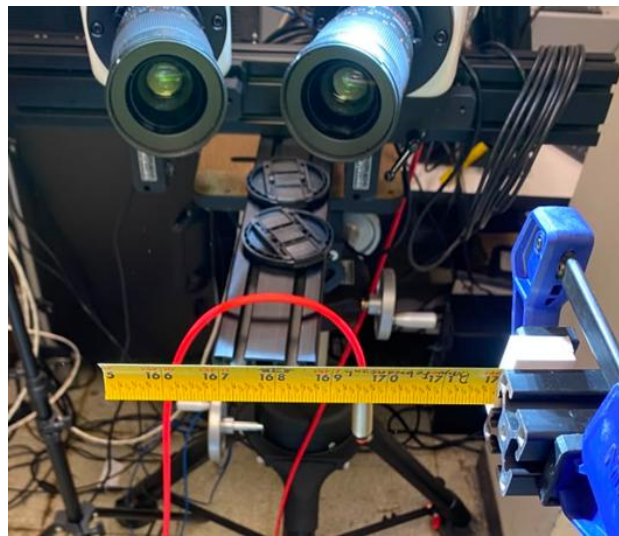


Figure 5. Vibration DIC setup.

Once the setup was prepared, the tape spring was excited by tapping it moderately with a screwdriver. This tap provided an impulse-like disturbance to the mounted tape spring, causing it to vibrate freely about its deployed equilibrium configuration. The purpose of this test was to measure the natural frequency content of the mounted tape spring system.

After the vibration recording was completed, an initial FFT analysis was performed directly within the DIC software. This internal DIC FFT result provided the first estimate of the dominant vibration frequencies from the measured displacement field. The same vibration dataset was then exported as a sequence of CSV files, with each CSV file corresponding to one recorded frame. These exported frame-by-frame data files were post-processed in MATLAB to perform a separate FFT analysis, allowing the experimental frequency response to be checked in more detail.

The MATLAB script processed the vibration data from frame 44 through frame 4573 using the file naming format `Vibration-%05d_0.csv`. The sampling frequency was defined as $F_s = 2000$

Hz to match the high-speed DIC capture rate. For each frame, the script read the reference point coordinates and the reconstructed 3D point positions from the exported CSV file. The initial frame was used as the reference geometry, and a small central region of the tape spring was selected for the vibration signal. This central region was chosen because points near the edges of the tape spring are more likely to contain DIC tracking noise or edge-related reconstruction errors. In the MATLAB code, the central region was selected by calculating the distance of each point from the median position of the initial point cloud and selecting the closest 5% of points, with a minimum of 20 reference points.

For each subsequent frame, the script matched the selected central-region points back to their corresponding reference locations using the DIC reference coordinates. The displacement signal was calculated from the change in the Z_p coordinate relative to the initial frame. Instead of using a single point, the displacement values from the matched central-region points were averaged to obtain one representative vibration signal for each frame. This averaging was used to reduce the influence of point-level noise while still preserving the dominant motion of the tape spring. If a frame contained too few valid matched points, the missing value was later filled by linear interpolation. The resulting time signal was then detrended before the FFT was applied.

The MATLAB analysis produced three main outputs. First, the averaged central-region displacement signal was plotted in the time domain to show the decaying vibration response after the tap. Second, the signal was multiplied by a Hann window and transformed into the frequency domain using the FFT. The resulting single-sided amplitude spectrum was plotted against frequency so that the dominant frequency peaks could be identified. Third, the number of successfully matched central-region points was plotted for each frame to check the tracking quality of the selected region throughout the test. The script also reported the dominant frequency, the frequency resolution, and the maximum detectable frequency, which is equal to the Nyquist frequency of $F_s/2 = 1000\text{Hz}$.

The MATLAB FFT result was then compared with the internal FFT result from the DIC software. Since both analyses used the same high-speed DIC vibration test, agreement between the two frequency results would indicate that the dominant peaks are not artifacts of a particular post-processing method. These experimental frequencies were later compared with the Abaqus frequency simulation to determine whether the measured peaks correspond to the predicted bending and twisting modes of the mounted tape spring.

3.3. Abaqus Frequency Analysis

An Abaqus frequency model was created to provide a numerical reference for the experimental vibration results obtained from the tapped DIC test. The purpose of this simulation was to predict the natural frequencies and mode shapes of the mounted tape spring system so that the MATLAB FFT and DIC software FFT results could be compared with a finite element model. This is conducted based on the lab instructions for the Abaqus frequency analysis, where the tape spring model is analyzed using a linear perturbation frequency step with the Lanczos eigensolver and the predicted mode shapes and frequencies are compared with the experimental vibration results.

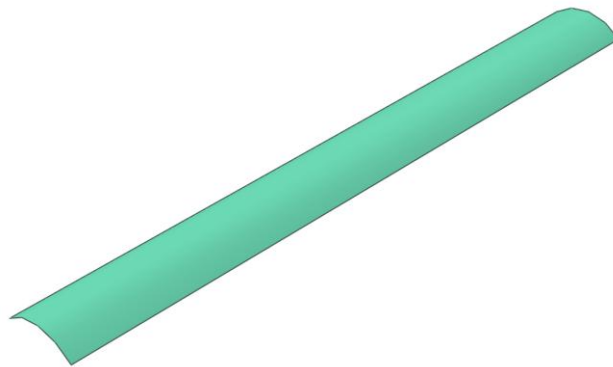


Figure 6. Tape spring Abaqus model.

The tape spring was first modeled in Abaqus as a three-dimensional shell structure, Figure 6. This was appropriate because the tape spring is very thin compared with its length and width, so its vibration behavior is dominated by shell-like bending and twisting deformation. The initial tape spring geometry followed the tape spring dimensions, with a thickness of 120 μm , length of 8 inches, and curved cross-section corresponding to the DIC tape radius of 16.5 mm. The material assigned to the tape spring was carbon steel with an elastic modulus of 200 GPa and Poisson's ratio of 0.3.

The 3D-printed clamp geometry was also included in the model. The top and bottom mount pieces were modeled as solid parts, representing the PLA fixture used to hold the tape spring during the physical vibration test. The PLA material properties were assigned using an elastic modulus of 3.6 GPa, Poisson's ratio of 0.3, and density of 1.24 g/cc. Including the mount in the model allowed the simulated boundary condition to more closely represent the experimental setup, where the tape spring was not fixed directly at the end but was held through the 3D-printed clamp.

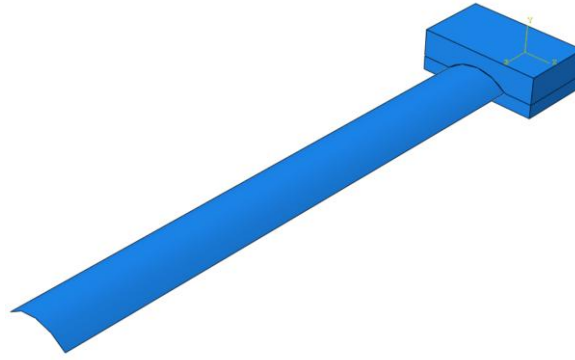


Figure 7. Tape spring clamped using the 3D printed PLA mount.

After the parts were created and material properties were assigned, the tape spring and the two mount components were assembled together, Figure 7. Two tie constraints were then defined to connect the tape spring to the clamp. The first tie constraint was applied between the top surface of the tape spring and the bottom surface of the upper mount. The second tie constraint was applied between the bottom surface of the tape spring and the top surface of the lower mount. These tie constraints were used to represent the bonded or tightly clamped contact between the tape spring and the 3D-printed mount, preventing relative motion between the connected surfaces during the frequency analysis.

The boundary condition was then applied to the mount to represent the experimental clamp that held the fixture to the stand. The mount was fixed so that its translational and rotational degrees of freedom were constrained. This simplified the real experimental support as a fixed boundary condition, allowing the free portion of the tape spring to vibrate relative to the clamped end. Although this does not capture every detail of the physical stand and clamp compliance, it provides a reasonable approximation of the mounted configuration used in the DIC vibration test.

The analysis step was defined as a frequency extraction step. A linear perturbation frequency procedure was used with the Lanczos eigensolver, and five eigenvalues were requested. The first several extracted modes were therefore available for inspection, although the main comparison in the report focuses on the first three vibration modes. The extracted mode shapes were used to identify whether each mode corresponded primarily to bending, twisting, or coupled bending-twisting motion.

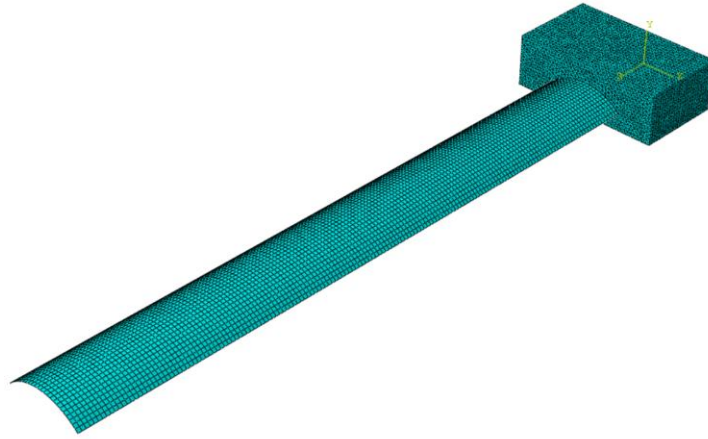


Figure 8. Meshed assembly.

Finally, the assembled model was meshed and submitted for analysis, Figure 8. After the job completed, the natural frequencies and corresponding mode shapes were obtained from the Abaqus results. These simulated frequencies were later compared with the dominant peaks from both the MATLAB FFT and the DIC software FFT.

3.4. Deployment

The deployment behavior of the tape spring was studied using a separate pull-and-release test. This test used a similar DIC camera and lighting setup as the radius and vibration measurements, but the purpose of the test was different. Instead of extracting a fitted radius or performing FFT analysis, the deployment test was used to observe the physical sequence of motion as the tape spring recovered from a bent configuration to its deployed state.

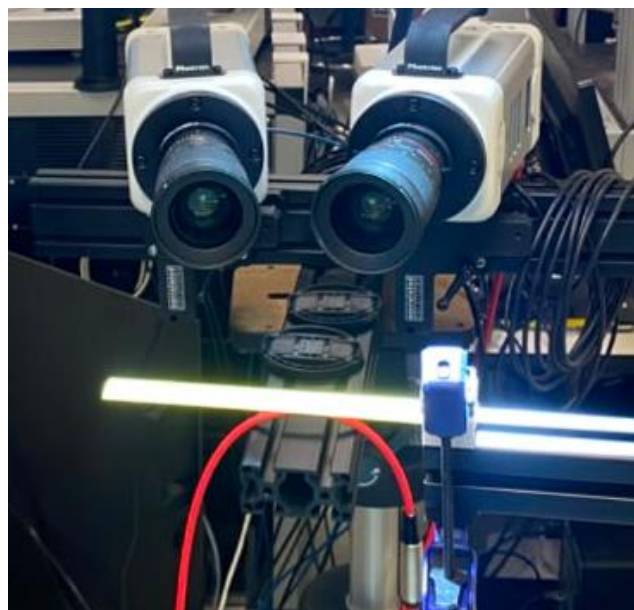


Figure 9. Deployment DIC setup.

For this test, the tape spring was positioned on its side relative to the cameras, Figure 9. This viewing direction was chosen so that the overall deployment motion could be seen clearly from the side, including the change in shape as the tape spring moved from the bent state toward the deployed state. As in the previous DIC measurements, the lighting and camera focus were adjusted before recording so that the tape spring remained visible during the high-speed capture. However, because the tape spring was viewed from the side, the speckle pattern on the surface was not as clearly visible as in the front-facing DIC measurements. As a result, this dataset was not used for detailed point-cloud reconstruction or quantitative displacement analysis.

The tape spring was initially bent manually away from its deployed configuration. This created a stored elastic energy state similar to the condition expected before deployment. Once the tape spring was released, it rapidly recovered toward its original deployed shape. The high-speed camera sequence captured this motion, including the initial bent configuration, the release and rapid recovery, the first fully deployed shape, and the subsequent vibration and damping.

Because the side-view speckle visibility was limited, only selected key frames from the deployment video were saved and analyzed. These frames were chosen to represent the main stages of the deployment process: the pre-release bent configuration, the rapid deployment stage, the near-deployed configuration, and the final damped state after the motion had mostly settled. The selected images were then used to qualitatively describe how the tape spring transitioned from stored elastic deformation to deployed equilibrium. This approach allowed the deployment behavior to be discussed in terms of visible shape change, recovery speed, overshoot, post-deployment oscillation, and damping, even though the side-view data were not suitable for full quantitative DIC post-processing.

3.5. Three-Point Bending

The quasi-static bending behavior of the tape spring was measured using a 3-point bending test. This experiment was performed twice, once for equal-sense bending and once for opposite-sense bending, so that the two bending directions could be compared directly. The purpose of this test was to obtain the experimental force-displacement response of the tape spring and then convert that response into a moment-angle curve for comparison with the analytical steady-state bending moment and the Abaqus bending simulation.

The test was performed using an Instron testing machine. The tape spring was placed horizontally on two lower supports, and the center loading fixture was aligned near the

midpoint of the span. The lower anvil span was set to 120 mm, the travel distance was 25 mm, and the loading rate was 10 mm/min. The measured force will be used to calculate the applied moment, while the measured crosshead displacement will be used to calculate the bending angle.



Figure 10. Equal-sense bending configuration.

For the equal-sense bending test, the tape spring was oriented so that the applied center displacement bent the tape spring in the same sense as its original curvature, Figure 10. For the opposite-sense bending test, the tape spring was flipped or reoriented so that the same 3-point bending setup imposed bending opposite to the original curvature, Figure 11. In both cases, the specimen was positioned carefully on the supports before testing so that the loading nose contacted the center region of the tape spring and the deformation occurred primarily within the intended bending configuration.



Figure 11. Opposite-sense bending configuration.

During each test, the Instron recorded the applied force and crosshead displacement. These raw force-displacement data were not compared directly with the analytical tape spring theory because the analytical model predicts steady-state bending moment rather than crosshead force. Therefore, the force and displacement data were post-processed into moment and bending angle using the relations introduced in Section 2.4.

The resulting moment-angle curves were then used to evaluate the nonlinear bending behavior of the tape spring. In particular, the plots were examined to determine whether the response showed the expected increase in moment followed by a steady-state moment region. The steady-state portions of the experimental curves were compared with the analytical equal-sense and opposite-sense moment predictions. The same experimental moment-angle data were also compared with the Abaqus bending simulation so that the measured response could be evaluated against both the simplified analytical model and the finite element model. Because the bending angle was calculated from crosshead displacement rather than measured directly from the tape spring shape, the angle should be interpreted as an effective bending angle.

3.6. Abaqus Bending Simulation

An Abaqus bending simulation was performed to provide a finite element comparison for the experimental 3-point bending results. The goal of this model was to reproduce the bending response of the tape spring numerically and obtain a simulated force-displacement or moment-angle response that could be compared with the Instron data and the analytical steady-state moment. The lab module recommends using a shorter tape measure model of 120 mm for better convergence, so the Abaqus bending model was simplified using a shortened tape spring rather than the full experimental length.

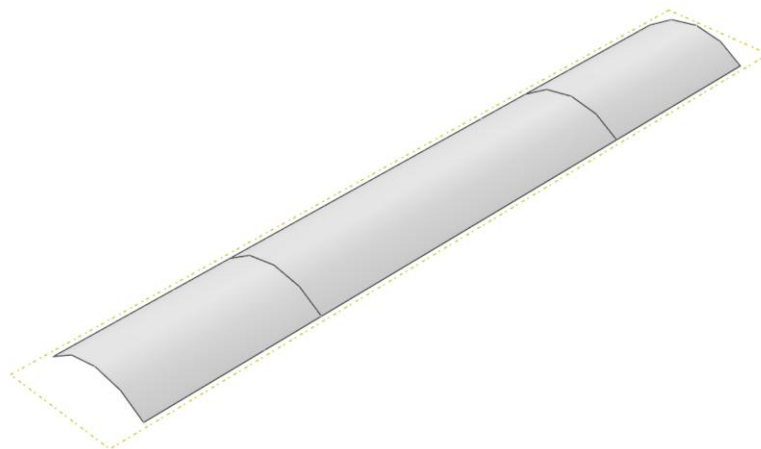


Figure 12. Tape spring model.

The tape spring was first modeled in Abaqus using the same curved cross-sectional geometry as the physical tape spring, Figure 12. Material properties were then assigned to the model, using carbon steel properties consistent with the analytical and frequency simulations. Because this model was intended to simplify the bending behavior, the loading nose and lower supports from the physical 3-point bending fixture were not modeled directly. Instead, the tape spring was bent using prescribed boundary conditions. This approach reduced contact complexity and helped the simulation converge more reliably.

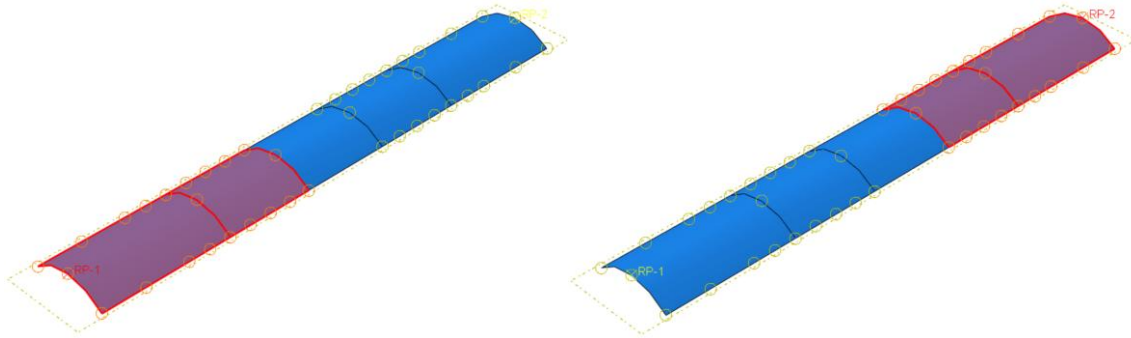


Figure 13. Tape spring partitioned constraints.

To control the deformation, the tape spring was partitioned, Figure 13. The left and right end sections were constrained so that the main bending deformation occurred in the middle section of the tape spring. This simplified the physical 3-point bending setup into a boundary-condition-driven bending problem, where the end regions were used to impose the rotation and displacement needed to bend the tape spring in a controlled way, Figure 14.

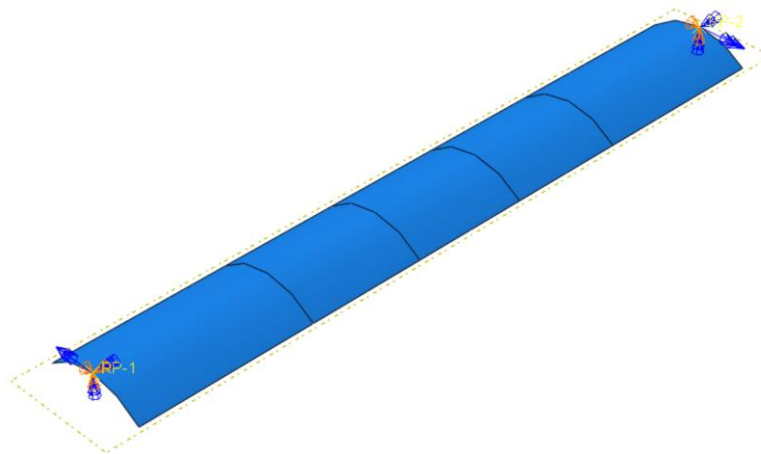


Figure 14. Tape spring model with boundary conditions.

The analysis step was defined as a static, general step. Rotation was applied through boundary conditions at the partitioned end regions. One end of the tape spring was constrained so that it could only rotate about the x -axis, while the other end was allowed to rotate about the x -axis

and displace in the z-direction. This was done so that the simulated deformation would resemble the bending shape produced during the 3-point bending experiment, while still allowing enough freedom for the model to deform without excessive artificial constraint.

Separate simulations were performed for equal-sense and opposite-sense bending. The bending direction was controlled by the sign of the applied rotation/displacement boundary condition. This allowed the two bending cases to be compared directly with the experimental equal-sense and opposite-sense moment-angle curves. After the boundary conditions and analysis step were defined, the model was meshed and the job was submitted. The output from the Abaqus simulation was obtained as a moment-angle response.

4. Results and Analysis

This section presents the experimental and numerical results obtained from the DIC measurements, vibration testing, deployment observation, and 3-point bending test. For each case, the experimental result is compared with the appropriate analytical or numerical reference so that the agreement and discrepancy between theory, experiment, and simulation can be evaluated.

The DIC radius measurements are first used to validate the analytical curvature prediction. The vibration response is then compared across the DIC software FFT, MATLAB FFT, and Abaqus frequency model. The deployment sequence is discussed qualitatively using selected image frames from the pull-and-release test. Finally, the 3-point bending moment-angle results are compared with the analytical steady-state moments and Abaqus bending simulation to evaluate the nonlinear bending behavior of the tape spring.

4.1. Radius of Curvature

The DIC radius measurements were processed in MATLAB by fitting the reconstructed point-cloud geometry of the tape spring to a cylindrical surface. This was done for the flat/deployed configuration, equal-sense bending configuration, and opposite-sense bending configuration. The measured shell and fitted cylinder plots are shown in Figure 15, 16, and 17. In each plot, the measured DIC surface is shown as a shell, the fitted cylinder is overlaid transparently, and the fitted cylinder axis is shown as a black line.

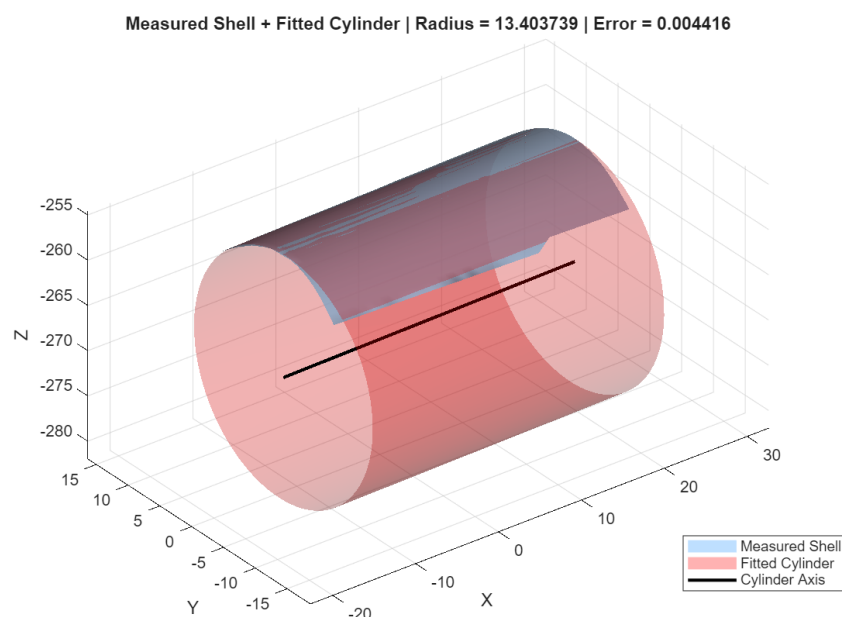


Figure 15. Flat fitted cylinder.

Measured Shell + Fitted Cylinder | Radius = 16.507726 | Error = 0.004620

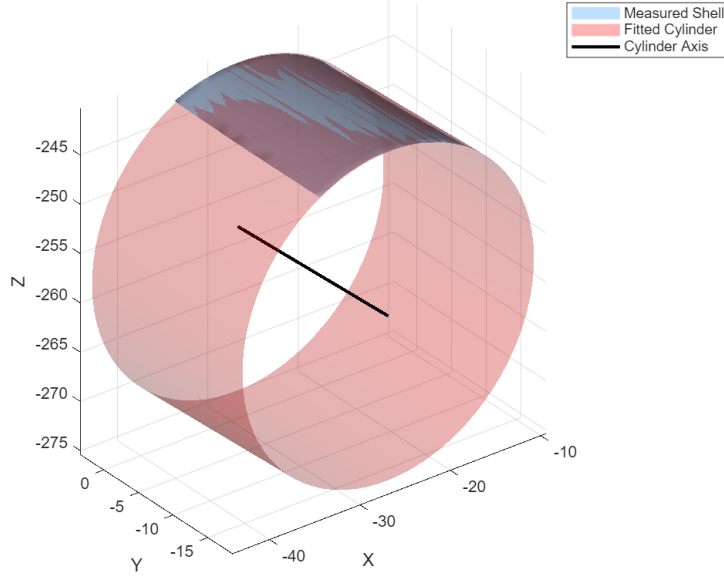


Figure 16. Equal-sense bending fitted cylinder.

Measured Shell + Fitted Cylinder | Radius = 15.878806 | Error = 0.003857

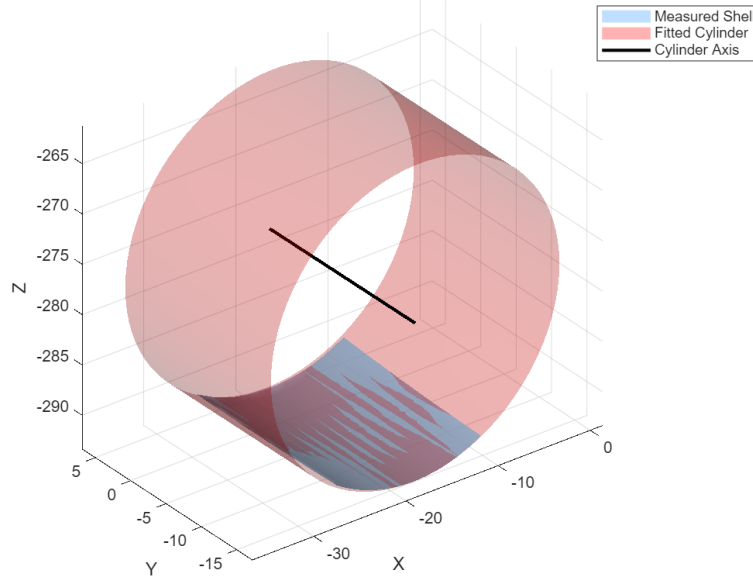


Figure 17. Opposite-sense bending fitted cylinder.

The fitted radius values are summarized in Table 1. The nominal reference radius was taken as $R = 16.25$ mm, which gives an analytical curvature of approximately 61.54 m^{-1} . The DIC-fitted radii were converted into curvature using

$$\kappa = \frac{1}{r}$$

with r converted from millimeters to meters.

Table 1. DIC radius compared to the reference radius.

Configuration	DIC Radius (mm)	DIC Curvature (m^{-1})	Error vs. 16.25 mm
Flat/deployed	13.4037	74.61	-17.52%
Equal-sense	16.5077	60.58	+1.58%
Opposite-sense	15.8788	62.98	-2.29%

Overall, the equal-sense and opposite-sense bending measurements were close to the analytical prediction. The equal-sense fitted radius was 16.5077 mm, while the opposite-sense fitted radius was 15.8788 mm. Both values are reasonably close to the nominal theoretical radius of 16.25 mm, supporting the analytical result from Section 2.1 that the bending radius of the localized tape spring hinge is expected to be approximately equal in magnitude to the original transverse radius of curvature. The corresponding curvatures, 60.58 m^{-1} and 62.98 m^{-1} , are also close to the analytical curvature of 61.54 m^{-1} .

The flat/deployed configuration showed a larger discrepancy, with a fitted radius of 13.4037 mm. This is noticeably smaller than the nominal reference radius. One likely reason is that the tape spring was not perfectly cylindrical in its undeformed state. In practice, the physical tape spring may have slight deviations from a perfectly round cross-section due to design choices, manufacturing variation, handling, or any residual deformation. Since the MATLAB fit reduces the measured surface to a single best-fit cylinder, any nonuniformity in the real cross-section can shift the fitted radius away from the nominal value. Therefore, the flat/deployed fitted radius should be interpreted as the best cylindrical approximation of the measured surface rather than as proof that the entire tape spring had a perfectly uniform radius.

The difference between the equal-sense and opposite-sense bending radii can also be explained by the side of the tape spring that was visible to the DIC system. The DIC measurements were taken from the same camera side for both bending directions. Because of this, the measured surface corresponded to the outer curvature in the equal-sense case but the inner curvature in the opposite-sense case. Since the tape spring has a finite thickness of $120 \mu\text{m}$, the inner and outer surfaces are not located at the same radial position. This means that even if the neutral-surface bending radius were identical for equal-sense and opposite-sense bending, the DIC-measured surface radius could differ slightly depending on whether the visible surface was on

the inside or outside of the bend. This thickness effect is small, but it contributes to the measured difference between the equal-sense and opposite-sense fitted radii.

The fitting errors were small for all three cases, indicating that the cylinder fits were visually and numerically reasonable for the measured point clouds. However, several sources of uncertainty remain. The tape spring was manually bent into the equal-sense and opposite-sense fixtures, so small differences in placement or alignment could affect the measured geometry. The 3D-printed mounts may also introduce slight constraint differences between configurations. In addition, DIC reconstruction quality depends on speckle quality, lighting, camera calibration, focus, and the visibility of the tracked surface. Finally, the cylinder-fitting method assumes that the measured portion of the tape spring can be represented by a perfect cylindrical surface, while the real tape spring may contain local flattening, twisting, or nonuniform curvature. Despite these limitations, the DIC results show that the bent configurations follow the expected theoretical radius trend reasonably well.

4.2. Vibration Frequency

The vibration response of the mounted tape spring was first analyzed using the internal FFT tool in the DIC software, since this function is built directly into the DIC post-processing workflow.

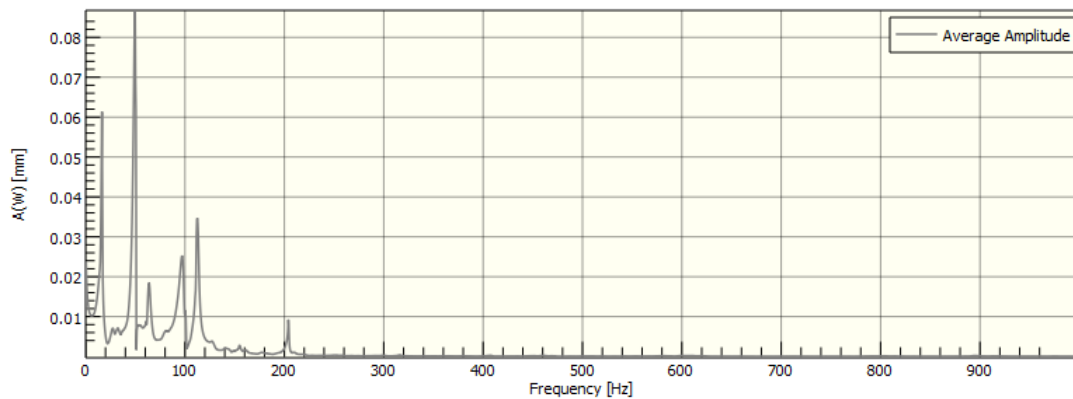


Figure 18. DIC software FFT analysis.

The resulting DIC FFT spectrum is shown in Figure 18. This plot provides the initial experimental estimate of the frequency content from the tapped vibration test. Since only the DIC FFT plot was available and the exact peak data were not exported, the DIC software frequencies reported in this section were estimated by interpolating from the plotted frequency axis. Therefore, these values should be treated as approximate rather than exact peak-picked frequencies.

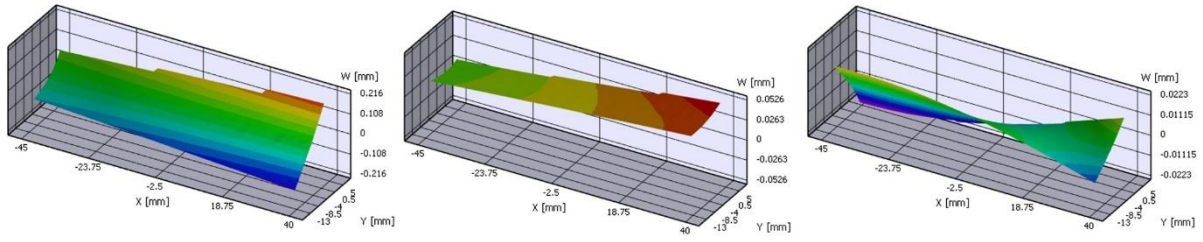


Figure 19. DIC software FFT modes; mode 1 (left), mode 2 (middle), mode 3 (right).

The DIC software FFT shows several noticeable peaks in the lower-frequency range. The first major peak occurs at approximately 50 Hz, while another strong peak appears near approximately 115 Hz. A weaker higher-frequency feature is also visible around approximately 205 Hz. These peaks indicate that the mounted tape spring response is dominated by lower-frequency vibration modes after being excited by the screwdriver tap, Figure 19.

The same vibration dataset was then exported as a sequence of CSV files and post-processed separately in MATLAB. The MATLAB FFT result is shown in Figure 20.

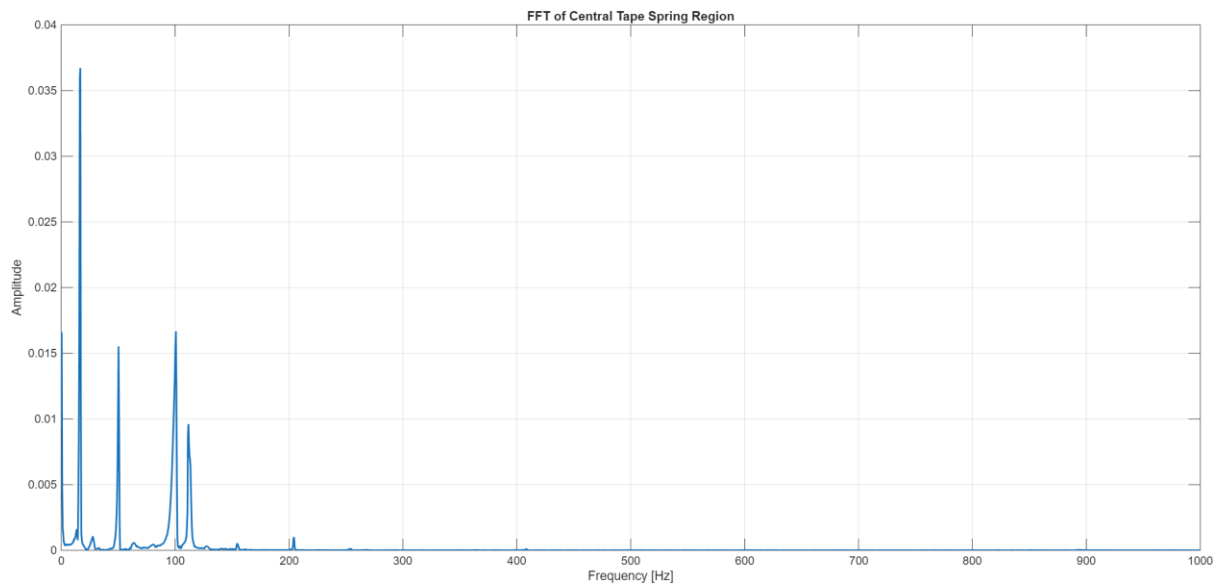


Figure 20. MATLAB FFT analysis.

The MATLAB spectrum shows the same general frequency behavior as the DIC software FFT, with the response concentrated mainly below approximately 250 Hz. The first clear peak occurs at 50.331 Hz, followed by a stronger peak at 111.70 Hz. A weaker higher-frequency peak is also visible at 204.415 Hz. The close agreement between the MATLAB FFT and the DIC software FFT suggests that the main peaks are not only artifacts of the DIC software's internal processing, but are also present when the data are independently processed from the exported frame-by-frame CSV files. Since the MATLAB FFT was obtained from an averaged central-

region displacement signal, the plotted peaks mainly represent the dominant vibration response of the middle portion of the tape spring.

Finally, the experimental frequency results were compared with the Abaqus frequency simulation. The Abaqus model predicted the first three natural frequencies as 62.746 Hz, 117.20 Hz, and 266.20 Hz. The first mode shape is primarily a twisting mode of the free portion of the tape spring. The second mode is a bending mode, oscillating up and down relative to the y-axis. The third mode shows a more complex deformation pattern, with stronger coupled bending-twisting behavior. These mode shapes are shown in Figure 21, 22, and 23 to provide the numerical reference for interpreting the experimental FFT peaks.

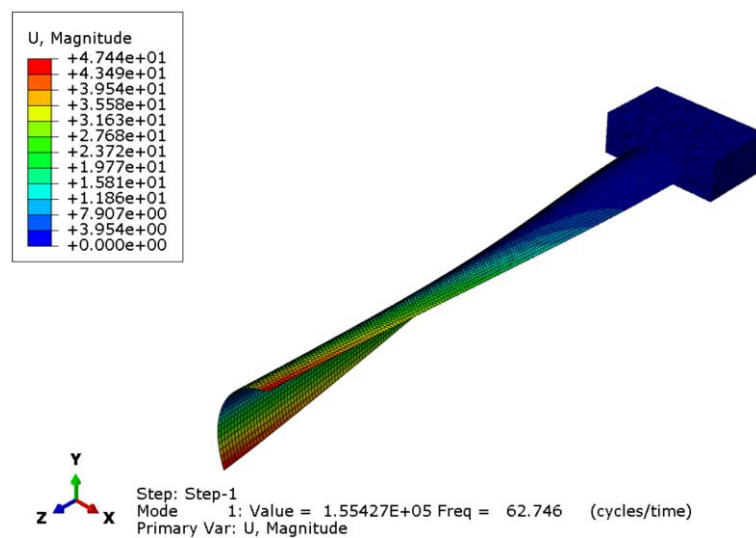


Figure 21. Abaqus first mode.

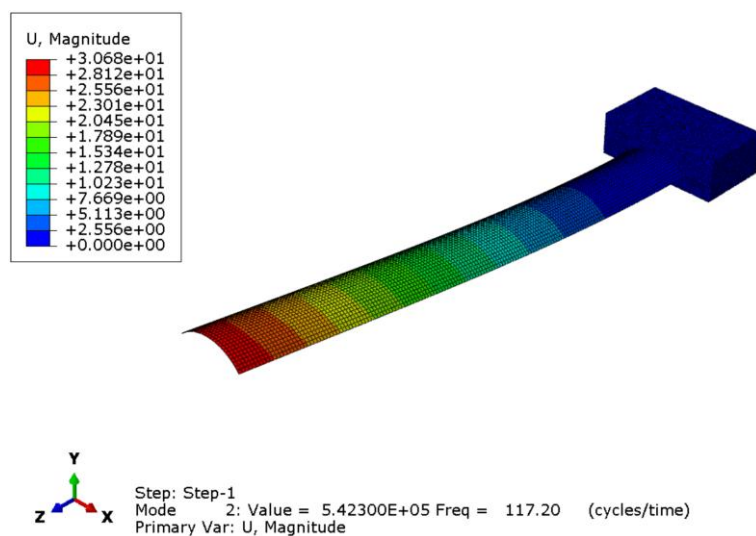


Figure 22. Abaqus second mode.

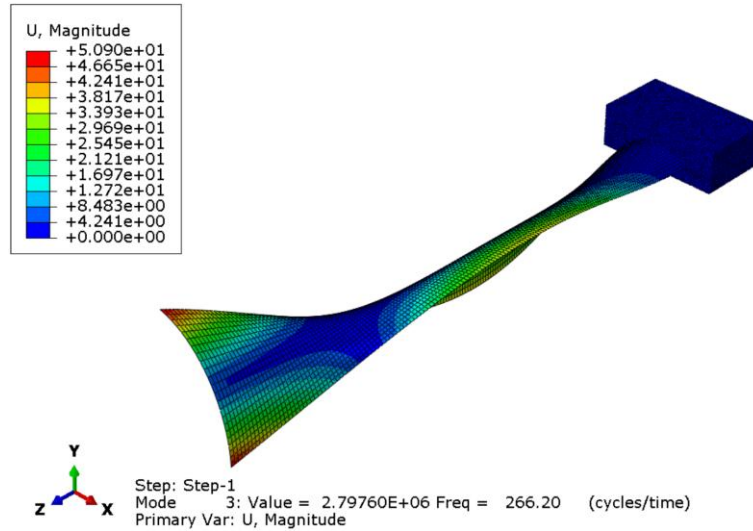


Figure 23. Abaqus third mode.

Table 2. Frequency Comparison.

Mode	DIC Software FFT Frequency	MATLAB FFT Frequency	Abaqus Frequency
1	~50 Hz	50.331 Hz	62.746 Hz
2	~115 Hz	111.70 Hz	117.20 Hz
3	~205 Hz	204.415 Hz	266.20 Hz

Table 2 shows that the DIC software FFT and MATLAB FFT are generally consistent with each other. Both experimental methods identify a first peak near 50 Hz, a second peak near 112–115 Hz, and a weaker higher-frequency peak near 204–205 Hz. The second experimental peak agrees most closely with the Abaqus second mode at 117.20 Hz, while the first and third experimental peaks are lower than the corresponding Abaqus predictions of 62.746 Hz and 266.20 Hz. Therefore, the experimental data and simulation capture the same general modal behavior, but the agreement is stronger for the second mode than for the first and third modes.

The discrepancies are expected because the Abaqus model does not perfectly capture the full dynamic behavior of the physical tape spring system. In the model, the tape spring was treated as carbon steel with idealized material properties. In the actual experiment, the tape spring surface was covered with paint and speckle markings for DIC tracking, which may slightly change the mass distribution and damping behavior. Although this effect is likely small, it can still shift the measured vibration response compared with an ideal carbon-steel model.

The mounting condition is another important source of difference. The Abaqus model included the PLA 3D-printed mount, but the physical experiment also included the larger stand and clamp system that held the mount in place. The stand may introduce additional compliance, looseness, or vibration that is not represented in the finite element model. As a result, the real boundary condition may be less rigid than the fixed constraint used in Abaqus. A less rigid support would generally reduce the measured natural frequencies, which is consistent with the experimental peaks being lower than the Abaqus predictions for the first and third modes.

Additional experimental uncertainties may also contribute to the mismatch. The tap excitation was applied manually, so the direction and strength of the input were not perfectly controlled. Some modes may have been excited more strongly than others depending on where and how the tape spring was tapped. The DIC frequency results can also be affected by tracking noise, limited speckle visibility, camera resolution, and the specific displacement component used for the FFT. In the MATLAB processing, the selected central region may emphasize some modes while underrepresenting others, especially if a mode shape has smaller displacement near the selected region.

Overall, the vibration results show that the DIC software FFT, MATLAB FFT, and Abaqus model identify broadly similar low-frequency dynamic behavior. The second mode shows the closest agreement between experiment and simulation, while the first and third modes show larger differences. This suggests that the Abaqus model captures the general modal behavior of the mounted tape spring, but the experimental system includes additional effects from paint, fixture compliance, stand dynamics, manual tapping, and DIC measurement uncertainty.

4.3. Deployment Behavior

Figure 24 shows selected frames from the pull-and-release deployment sequence of the tape spring. The frames are arranged chronologically to show how the tape spring transitions from a manually bent configuration to its final deployed state. Since the tape spring was viewed from the side and the speckle pattern was not clearly visible throughout the entire motion, this sequence was used for qualitative interpretation rather than full DIC point-cloud or displacement analysis. However, the visible deformation still provides useful insight into the dominant motion types excited during deployment.

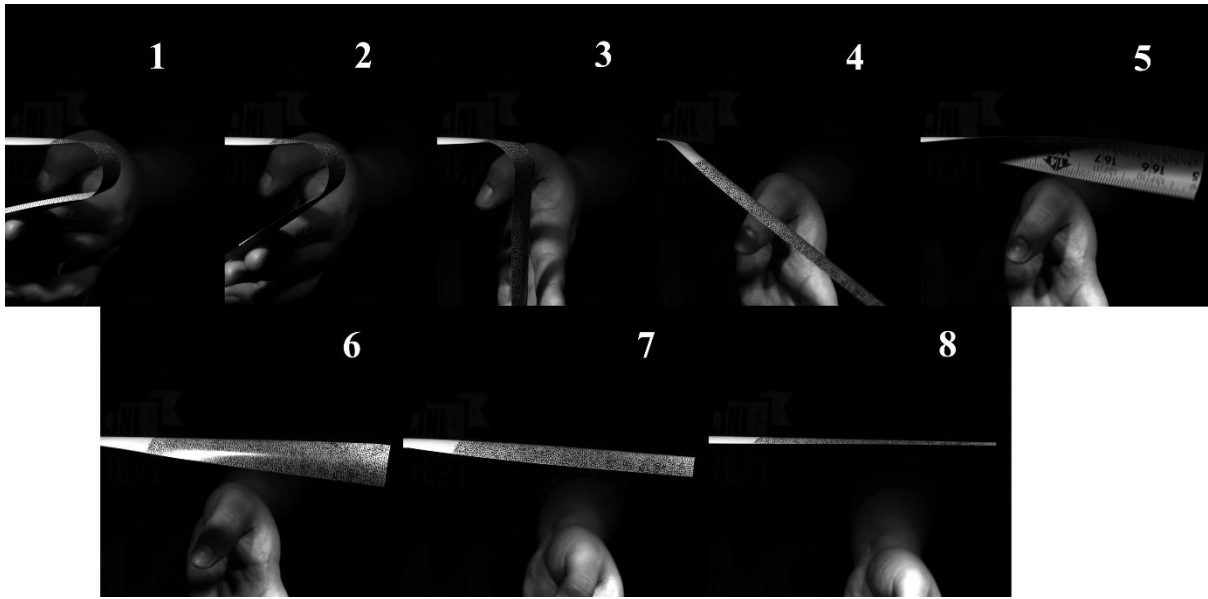


Figure 24. 8 frames of deployment sequence.

In Frame 1, the tape spring is shown in the pre-release bent configuration. At this stage, the structure is manually held away from its natural deployed shape, so elastic strain energy is stored through bending and local curvature change. The deformation appears primarily bending-dominated, since the tape spring is held in a curved configuration before release. This frame represents the initial stored-energy condition before deployment begins.

Frames 2 and 3 show the early release and rapid deployment stage. After the tape spring is released, the stored elastic energy is converted into kinetic energy, causing the structure to move quickly away from the bent configuration. The tape spring begins recovering its original shape, but the motion is not a simple rigid-body rotation. Instead, the changing orientation and visible curvature indicate that elastic recovery occurs along the length of the tape spring. This stage is mainly associated with bending recovery, although the slight change in orientation suggests that twisting may also begin to be excited.

In Frame 4, the tape spring has passed through an intermediate deployed direction and appears to be moving with significant momentum. This frame suggests that the release is dynamic, with the tape spring continuing past the position it would occupy under a purely quasi-static recovery. The motion also appears slightly off-axis, which may be caused by the manual release, the initial bending direction, or bending-twisting coupling.

Frames 5 and 6 show the most apparent twisting-type motion. In these frames, the tape spring is close to its deployed lengthwise configuration, but the visible orientation of the strip changes across the span. The cross-section no longer appears to move only up and down in a planar

bending motion; instead, the tape spring appears to rotate about its length. This is consistent with a torsional or coupled bending-twisting response being excited during deployment. This observation also agrees qualitatively with the modal interpretation from the vibration analysis, where the tape spring response was not expected to be purely bending due to its curved cross-section and clamped boundary condition.

Frame 7 shows the post-deployment oscillation stage. The tape spring is close to its deployed shape but still appears to be vibrating about the equilibrium position. At this point, the motion appears to contain a combination of bending and residual twisting. This behavior is expected because the release process excites multiple deformation modes at once rather than only a single pure mode.

Finally, Frame 8 shows the damped steady-state configuration. By this point, the visible motion has mostly decayed, and the tape spring has returned to a stable deployed shape. The reduction in motion between the earlier frames and this final frame indicates that energy has been dissipated through material damping, air resistance, clamp interaction, and other small losses in the experimental setup.

Overall, the image sequence shows the expected stages of tape spring deployment: pre-release bending, rapid elastic recovery, overshoot, post-deployment oscillation, and damping to a final deployed state. The sequence also shows that the deployment is not purely planar. The early frames are dominated by bending recovery, while the later frames, especially Frames 5 and 6, show visible twisting or coupled bending-twisting motion. This supports the interpretation that deployment of the tape spring excites multiple dynamic deformation modes before the structure settles into its final deployed configuration.

4.4. Three Point Bending

The 3-point bending results were converted from force-displacement data into moment-angle form using the relations described in Section 2.4. The experimental curve was then compared with the Abaqus bending simulation and the analytical steady-state moment predictions, as shown in Figure 25.

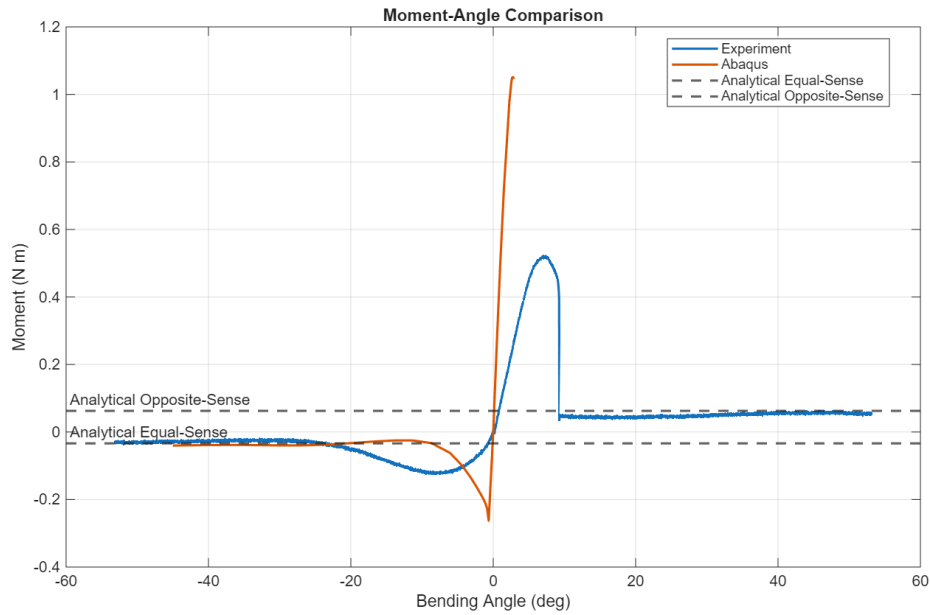


Figure 25. Moment-angle curve for experiment, simulation, and analytical calculation.

The experimental curve shows a strongly nonlinear bending response. On the negative-angle side, the moment initially remains close to the analytical equal-sense steady-state value, then becomes more negative as the bending angle approaches the transition region near the origin. This portion of the curve is reasonably consistent with the analytical expectation that the equal-sense bending moment should be smaller in magnitude than the opposite-sense moment. The experimental response does not stay perfectly constant, however, which suggests that the physical test includes additional effects such as contact with the supports, fixture compliance, local curvature changes, and possible sliding or seating of the tape spring during loading.

On the positive-angle side, the experimental moment increases sharply and reaches a large peak of approximately 0.52 N·m. After this peak, the moment drops suddenly to a lower plateau region of approximately 0.04–0.06 N·m, which is much closer to the analytical opposite-sense steady-state value. This behavior suggests that the large positive peak should not be interpreted as the steady-state tape spring moment. Instead, it is likely associated with a local instability, snap-through, contact change, or sudden release of stored elastic energy during the physical bending test. After this transition, the post-drop plateau provides a more meaningful comparison to the analytical steady-state prediction.

The Abaqus simulation captures the nonlinear trend of the bending response, but the agreement is not exact. The negative branch follows the same general direction as the experimental negative branch and remains near the analytical equal-sense steady-state level for much of the response. In the positive direction, the Abaqus moment rises rapidly and reaches a much larger

value than the analytical steady-state prediction. The simulation stopped at this peak because the solution failed to converge. This likely occurred because the model reached an instability point where a large amount of stored elastic energy was released over a very small change in deformation, making the static solution difficult for Abaqus to continue.

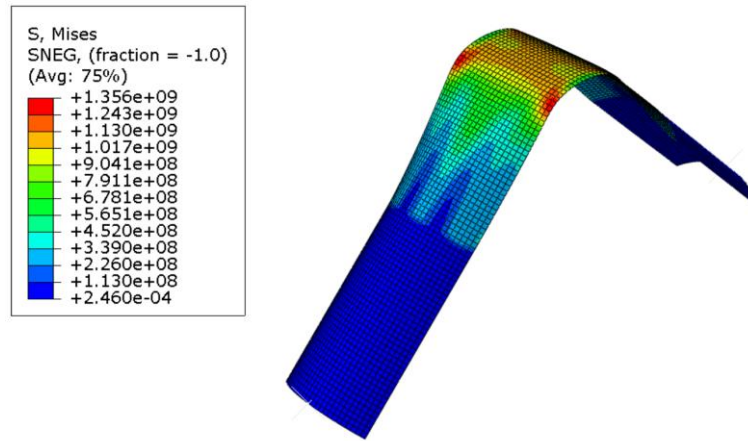


Figure 26. Equal-sense bending Abaqus simulation.

The Abaqus stress contours also show that the bending deformation is highly localized. In both bending cases, the largest stresses appear near the hinge-like region where the tape spring undergoes the strongest curvature change, Figure 26 and 27. This supports the idea that the tape spring does not bend uniformly along its length, but instead forms a localized bending region. The higher stress concentration in the more severe bending case also helps explain why the Abaqus model had difficulty converging near the peak moment. Once the deformation becomes highly localized, small changes in angle can cause large changes in moment and stress, which can lead to numerical instability in a static general analysis.

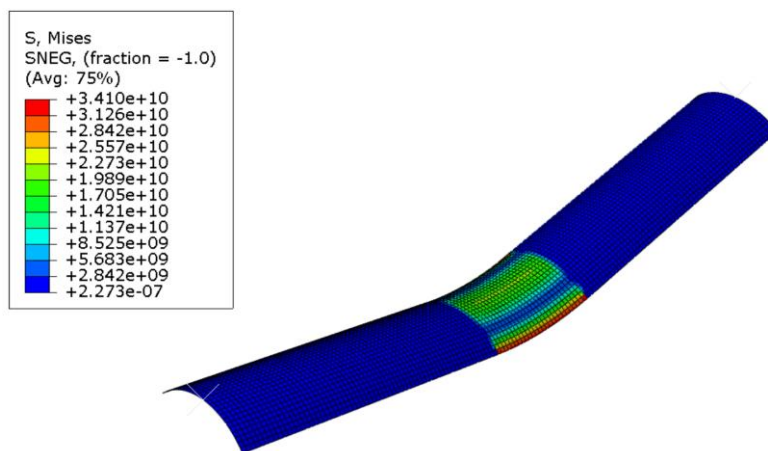


Figure 27. Opposite-sense bending Abaqus simulation.

Several factors explain the differences between the experiment, Abaqus, and analytical theory. The analytical model predicts ideal steady-state moments for a simplified tape spring geometry, so it does not capture transient peaks, support contact, friction, local slipping, or fixture compliance. The experimental 3-point bending test includes these effects because the tape spring physically contacts the loading nose and lower supports while deforming. The Abaqus model is also simplified because the bending was imposed through boundary conditions rather than by explicitly modeling the full 3-point bending fixture and contact interactions. As a result, the Abaqus curve is useful for comparing the general nonlinear bending trend, but it does not reproduce every detail of the physical experiment.

Overall, the 3-point bending results show that the tape spring exhibits the expected nonlinear moment-angle behavior. The equal-sense side has a smaller steady-state moment magnitude, while the opposite-sense side produces a larger resistance and a pronounced instability before reaching the lower steady-state region. The analytical steady-state values provide a reasonable reference for the post-transition plateau regions, while the experimental and Abaqus curves show the additional nonlinear effects that occur during the actual bending process.

5. Conclusion

This report investigated the geometric, dynamic, deployment, and bending behavior of a carbon-steel tape spring hinge using analytical theory, DIC measurements, MATLAB post-processing, Instron testing, and Abaqus simulations. The analytical model predicted that the localized bending radius should be approximately equal to the original transverse radius of curvature of the tape spring. This was generally supported by the DIC cylinder-fitting results for the equal-sense and opposite-sense bending configurations, which produced fitted radii of 16.5077 mm and 15.8788 mm, respectively, compared with the nominal reference radius of 16.25 mm. The flat/deployed configuration showed a larger discrepancy, with a fitted radius of 13.4037 mm, likely because the physical tape spring was not perfectly cylindrical in its undeformed state. Additional differences between the equal-sense and opposite-sense radii were attributed to the DIC system measuring different visible surfaces of the tape spring depending on the bending direction, as well as fixture alignment, DIC reconstruction uncertainty, and cylinder-fitting assumptions.

The vibration analysis showed that the DIC software FFT and MATLAB FFT produced consistent dominant frequency peaks. The DIC software FFT identified approximate peaks near 50 Hz, 115 Hz, and 205 Hz, while the MATLAB FFT produced corresponding peaks at 50.331 Hz, 111.70 Hz, and 204.415 Hz. These results indicate that the main frequency peaks were not artifacts of a single processing method. The Abaqus frequency model predicted the first three modes at 62.746 Hz, 117.20 Hz, and 266.20 Hz. The second mode showed the closest agreement with the experimental results, while the first and third modes showed larger discrepancies. These differences are reasonable because the Abaqus model used idealized material properties and boundary conditions, while the physical system included paint and speckle mass, the PLA mount, the clamp stand, manual tapping variation, and DIC tracking uncertainty.

The deployment sequence showed that the tape spring recovery was dynamic rather than quasi-static. The selected image frames showed a transition from a pre-release bent configuration to rapid elastic recovery, overshoot, post-deployment oscillation, and final damping to a steady deployed state. The motion was not purely planar; the later frames showed visible twisting or coupled bending-twisting behavior, especially after the main deployment motion. This agrees qualitatively with the vibration results, where the tape spring response included both bending and twisting-type modes.

The 3-point bending results also showed the expected nonlinear moment-angle behavior. The equal-sense side had a smaller steady-state moment magnitude, while the opposite-sense side produced a larger resistance. The experimental positive branch reached a large peak of approximately $0.52 \text{ N}\cdot\text{m}$ before dropping to a lower plateau near $0.04\text{--}0.06 \text{ N}\cdot\text{m}$, which was closer to the analytical opposite-sense steady-state moment. The Abaqus bending simulation captured the general nonlinear trend but did not reproduce the experiment exactly. The opposite-sense simulation stopped near the peak because the model likely reached an instability point where the static solution could no longer converge. The differences between the analytical, experimental, and Abaqus results are mainly attributed to the simplified analytical assumptions, the absence of full contact modeling in the Abaqus bending simulation, fixture compliance, support friction, local slipping, and the complex localized hinge deformation of the tape spring.

Overall, the tape spring behavior was captured well at the qualitative level. The DIC curvature measurements supported the theoretical prediction for the bent configurations, the FFT analyses identified consistent experimental vibration frequencies, the deployment images showed the expected recovery and damping behavior, and the bending tests demonstrated the nonlinear moment-angle response expected from a curved thin-shell structure. The remaining discrepancies are reasonable given the sensitivity of the tape spring to boundary conditions, measurement method, fixture interaction, and geometric nonlinearity. Therefore, the combined analytical, experimental, and numerical results provide a coherent understanding of the tape spring hinge behavior while also showing the limitations of idealized models when compared with real experimental behavior.

Appendix

DIC

```
close all
clear all
clc
% Read x, y, z data from Excel
data = readmatrix('Flat.xlsx');

% Keep only valid numeric rows and first 3 columns
data = data(:,1:3);
data = data(all(isfinite(data),2), :);

% Create point cloud
ptCloud = pointCloud(data);

% Reference axis estimate using PCA
[coeff, score, ~] = pca(data);
refVector = coeff(:,1);

% Cylinder fit
maxDistance = 0.01; % adjust if needed
[model, inlierIdx, outlierIdx, meanError] = pcfitcylinder(ptCloud, maxDistance,
refVector);

% Extract cylinder info
radius = model.Radius;
center = model.Center;
axisDir = model.Orientation(:);
axisDir = axisDir / norm(axisDir);

fprintf('Cylinder radius      = %.6f\n', radius);
fprintf('Mean fitting error   = %.6f\n', meanError);
fprintf('Number of inliers      = %d\n', numel(inlierIdx));
fprintf('Total data points      = %d\n', size(data,1));

%% Use inlier points to define cylinder plotting length
inlierPts = data(inlierIdx, :);
t = (inlierPts - center) * axisDir;
tMin = min(t);
tMax = max(t);
height = tMax - tMin;

%% Build fitted cylinder surface
nTheta = 80;
[xc, yc, zc] = cylinder(radius, nTheta);
zc = zc * height + tMin;

% Build basis perpendicular to cylinder axis
if abs(dot(axisDir, [1;0;0])) < 0.9
    temp = [1;0;0];
else
    temp = [0;1;0];
end

u = cross(axisDir, temp); u = u / norm(u);
v = cross(axisDir, u);   v = v / norm(v);
```

```

% Transform cylinder surface into global coordinates
Xcyl = center(1) + u(1)*xc + v(1)*yc + axisDir(1)*zc;
Ycyl = center(2) + u(2)*xc + v(2)*yc + axisDir(2)*zc;
Zcyl = center(3) + u(3)*xc + v(3)*yc + axisDir(3)*zc;

%% Build shell/surface representation of measured geometry
% Project points into a 2D PCA plane, triangulate there, then map back in 3D
uv = score(:,1:2);
tri = delaunay(uv(:,1), uv(:,2));

% Remove overly large triangles so the shell looks cleaner
P1 = data(tri(:,1), :);
P2 = data(tri(:,2), :);
P3 = data(tri(:,3), :);

e1 = vecnorm(P1 - P2, 2, 2);
e2 = vecnorm(P2 - P3, 2, 2);
e3 = vecnorm(P3 - P1, 2, 2);

edgeMedian = median([e1; e2; e3]);
keepTri = (e1 < 3*edgeMedian) & (e2 < 3*edgeMedian) & (e3 < 3*edgeMedian);
tri = tri(keepTri, :);

%% Plot
figure;
hold on;

% Measured geometry as shell
trisurf(tri, data(:,1), data(:,2), data(:,3), ...
    'FaceColor', [0.7 0.85 1.0], ...
    'EdgeColor', 'none', ...
    'FaceAlpha', 0.85, ...
    'DisplayName', 'Measured Shell');

% Fitted cylinder
surf(Xcyl, Ycyl, Zcyl, ...
    'FaceColor', [1 0 0], ...
    'EdgeColor', 'none', ...
    'FaceAlpha', 0.30, ...
    'DisplayName', 'Fitted Cylinder');

% Cylinder axis
axisStart = center + tMin * axisDir;
axisEnd = center + tMax * axisDir;
plot3([axisStart(1) axisEnd(1)], ...
    [axisStart(2) axisEnd(2)], ...
    [axisStart(3) axisEnd(3)], ...
    'k-', 'LineWidth', 2, ...
    'DisplayName', 'Cylinder Axis');

xlabel('X');
ylabel('Y');
zlabel('Z');
title(sprintf('Measured Shell + Fitted Cylinder | Radius = %.6f | Error = %.6f',
    radius, meanError));

axis equal;
grid on;

```

```

view(3);
camlight headlight;
lighting gouraud;
legend('Location','best');

close all
clear all
clc
% Read x, y, z data from Excel
data = readmatrix('Equal.xlsx');

% Keep only valid numeric rows and first 3 columns
data = data(:,1:3);
data = data(all(isfinite(data),2), :);

% Create point cloud
ptCloud = pointCloud(data);

% Reference axis estimate using PCA
[coeff, score, ~] = pca(data);
refVector = coeff(:,1);

% Cylinder fit
maxDistance = 0.01; % adjust if needed
[model, inlierIdx, outlierIdx, meanError] = pcfitcylinder(ptCloud, maxDistance,
refVector);

% Extract cylinder info
radius = model.Radius;
center = model.Center;
axisDir = model.Orientation(:);
axisDir = axisDir / norm(axisDir);

fprintf('Cylinder radius      = %.6f\n', radius);
fprintf('Mean fitting error   = %.6f\n', meanError);
fprintf('Number of inliers      = %d\n', numel(inlierIdx));
fprintf('Total data points      = %d\n', size(data,1));

%% Use inlier points to define cylinder plotting length
inlierPts = data(inlierIdx, :);
t = (inlierPts - center) * axisDir;
tMin = min(t);
tMax = max(t);
height = tMax - tMin;

%% Build fitted cylinder surface
nTheta = 80;
[xc, yc, zc] = cylinder(radius, nTheta);
zc = zc * height + tMin;

% Build basis perpendicular to cylinder axis
if abs(dot(axisDir, [1;0;0])) < 0.9
    temp = [1;0;0];
else
    temp = [0;1;0];
end

u = cross(axisDir, temp); u = u / norm(u);

```

```

v = cross(axisDir, u);    v = v / norm(v);

% Transform cylinder surface into global coordinates
Xcyl = center(1) + u(1)*xc + v(1)*yc + axisDir(1)*zc;
Ycyl = center(2) + u(2)*xc + v(2)*yc + axisDir(2)*zc;
Zcyl = center(3) + u(3)*xc + v(3)*yc + axisDir(3)*zc;

%% Build shell/surface representation of measured geometry
% Project points into a 2D PCA plane, triangulate there, then map back in 3D
uv = score(:,1:2);
tri = delaunay(uv(:,1), uv(:,2));

% Remove overly large triangles so the shell looks cleaner
P1 = data(tri(:,1), :);
P2 = data(tri(:,2), :);
P3 = data(tri(:,3), :);

e1 = vecnorm(P1 - P2, 2, 2);
e2 = vecnorm(P2 - P3, 2, 2);
e3 = vecnorm(P3 - P1, 2, 2);

edgeMedian = median([e1; e2; e3]);
keepTri = (e1 < 3*edgeMedian) & (e2 < 3*edgeMedian) & (e3 < 3*edgeMedian);
tri = tri(keepTri, :);

%% Plot
figure;
hold on;

% Measured geometry as shell
trisurf(tri, data(:,1), data(:,2), data(:,3), ...
    'FaceColor', [0.7 0.85 1.0], ...
    'EdgeColor', 'none', ...
    'FaceAlpha', 0.85, ...
    'DisplayName', 'Measured Shell');

% Fitted cylinder
surf(Xcyl, Ycyl, Zcyl, ...
    'FaceColor', [1 0 0], ...
    'EdgeColor', 'none', ...
    'FaceAlpha', 0.30, ...
    'DisplayName', 'Fitted Cylinder');

% Cylinder axis
axisStart = center + tMin * axisDir;
axisEnd    = center + tMax * axisDir;
plot3([axisStart(1) axisEnd(1)], ...
    [axisStart(2) axisEnd(2)], ...
    [axisStart(3) axisEnd(3)], ...
    'k-', 'LineWidth', 2, ...
    'DisplayName', 'Cylinder Axis');

xlabel('X');
ylabel('Y');
zlabel('Z');
title(sprintf('Measured Shell + Fitted Cylinder | Radius = %.6f | Error = %.6f',
    radius, meanError));

axis equal;

```

```

grid on;
view(3);
camlight headlight;
lighting gouraud;
legend('Location','best');

close all
clear all
clc
% Read x, y, z data from Excel
data = readmatrix('Opposite.xlsx');

% Keep only valid numeric rows and first 3 columns
data = data(:,1:3);
data = data(all(isfinite(data),2), :);

% Create point cloud
ptCloud = pointCloud(data);

% Reference axis estimate using PCA
[coeff, score, ~] = pca(data);
refVector = coeff(:,1);

% Cylinder fit
maxDistance = 0.01; % adjust if needed
[model, inlierIdx, outlierIdx, meanError] = pcfitcylinder(ptCloud, maxDistance,
refVector);

% Extract cylinder info
radius = model.Radius;
center = model.Center;
axisDir = model.Orientation(:);
axisDir = axisDir / norm(axisDir);

fprintf('Cylinder radius      = %.6f\n', radius);
fprintf('Mean fitting error   = %.6f\n', meanError);
fprintf('Number of inliers      = %d\n', numel(inlierIdx));
fprintf('Total data points      = %d\n', size(data,1));

%% Use inlier points to define cylinder plotting length
inlierPts = data(inlierIdx, :);
t = (inlierPts - center) * axisDir;
tMin = min(t);
tMax = max(t);
height = tMax - tMin;

%% Build fitted cylinder surface
nTheta = 80;
[xc, yc, zc] = cylinder(radius, nTheta);
zc = zc * height + tMin;

% Build basis perpendicular to cylinder axis
if abs(dot(axisDir, [1;0;0])) < 0.9
    temp = [1;0;0];
else
    temp = [0;1;0];
end

```

```

u = cross(axisDir, temp); u = u / norm(u);
v = cross(axisDir, u); v = v / norm(v);

% Transform cylinder surface into global coordinates
Xcyl = center(1) + u(1)*xc + v(1)*yc + axisDir(1)*zc;
Ycyl = center(2) + u(2)*xc + v(2)*yc + axisDir(2)*zc;
Zcyl = center(3) + u(3)*xc + v(3)*yc + axisDir(3)*zc;

%% Build shell/surface representation of measured geometry
% Project points into a 2D PCA plane, triangulate there, then map back in 3D
uv = score(:,1:2);
tri = delaunay(uv(:,1), uv(:,2));

% Remove overly large triangles so the shell looks cleaner
P1 = data(tri(:,1), :);
P2 = data(tri(:,2), :);
P3 = data(tri(:,3), :);

e1 = vecnorm(P1 - P2, 2, 2);
e2 = vecnorm(P2 - P3, 2, 2);
e3 = vecnorm(P3 - P1, 2, 2);

edgeMedian = median([e1; e2; e3]);
keepTri = (e1 < 3*edgeMedian) & (e2 < 3*edgeMedian) & (e3 < 3*edgeMedian);
tri = tri(keepTri, :);

%% Plot
figure;
hold on;

% Measured geometry as shell
trisurf(tri, data(:,1), data(:,2), data(:,3), ...
    'FaceColor', [0.7 0.85 1.0], ...
    'EdgeColor', 'none', ...
    'FaceAlpha', 0.85, ...
    'DisplayName', 'Measured Shell');

% Fitted cylinder
surf(Xcyl, Ycyl, Zcyl, ...
    'FaceColor', [1 0 0], ...
    'EdgeColor', 'none', ...
    'FaceAlpha', 0.30, ...
    'DisplayName', 'Fitted Cylinder');

% Cylinder axis
axisStart = center + tMin * axisDir;
axisEnd = center + tMax * axisDir;
plot3([axisStart(1) axisEnd(1)], ...
    [axisStart(2) axisEnd(2)], ...
    [axisStart(3) axisEnd(3)], ...
    'k-', 'LineWidth', 2, ...
    'DisplayName', 'Cylinder Axis');

xlabel('X');
ylabel('Y');
zlabel('Z');
title(sprintf('Measured Shell + Fitted Cylinder | Radius = %.6f | Error = %.6f',
    radius, meanError));

```

```

axis equal;
grid on;
view(3);
camlight headlight;
lighting gouraud;
legend('Location','best');

```

FFT

```

clear; clc; close all

dataFolder = pwd;

firstFrame = 44;
lastFrame = 4573;

Fs = 2000;

files = strings(0);

for i = firstFrame:lastFrame
    fname = fullfile(dataFolder, sprintf("Vibration-%05d_0.csv", i));
    if isfile(fname)
        files(end+1) = fname;
    end
end

nFrames = numel(files);
fprintf("Number of frames found: %d\n", nFrames)

A0 = readmatrix(files(1), "NumHeaderLines", 1);

qref0 = A0(:,20);
rref0 = A0(:,21);
xyz0 = A0(:,22:24);

valid0 = all(isfinite([qref0 rref0 xyz0]),2);

qref0 = qref0(valid0);
rref0 = rref0(valid0);
xyz0 = xyz0(valid0,:);

x0 = xyz0(:,1);
y0 = xyz0(:,2);

centerXY = median([x0 y0], 1, "omitnan");

spanXY = range([x0 y0], 1);
spanXY(spanXY == 0) = 1;

distFromCenter = sqrt(sum((((x0 y0] - centerXY)./spanXY).^2, 2));

centralFraction = 0.05;

[~, order] = sort(distFromCenter);

nSelect = max(20, round(centralFraction*numel(x0)));

```

```

roiRef = order(1:nSelect);

qROI = qref0(roiRef);
rROI = rref0(roiRef);
zROI0 = xyz0(roiRef,3);

fprintf("Number of reference points in central ROI: %d\n", numel(roiRef))

sig = nan(nFrames,1);
nMatched = zeros(nFrames,1);

tol = 1e-6;

for k = 1:nFrames
    A = readmatrix(files(k), "NumHeaderLines", 1);

    qref = A(:,20);
    rref = A(:,21);
    xyz = A(:,22:24);

    valid = all(isfinite([qref rref xyz]),2);

    qref = qref(valid);
    rref = rref(valid);
    xyz = xyz(valid,:);

    dz = nan(numel(roiRef),1);

    for j = 1:numel(roiRef)
        dist = hypot(qref - qROI(j), rref - rROI(j));
        [dmin, idxMin] = min(dist);

        if dmin < tol
            dz(j) = xyz(idxMin,3) - zROI0(j);
        end
    end

    nMatched(k) = sum(isfinite(dz));

    if nMatched(k) >= 0.5*numel(roiRef)
        sig(k) = mean(dz, "omitnan");
    end
end

if any(isnan(sig))
    fprintf("Interpolating %d missing/weak frames.\n", sum(isnan(sig)))
    sig = fillmissing(sig, "linear");
end

sig = detrend(sig);

N = length(sig);
t = (0:N-1)'/Fs;

figure
plot(t, sig, "LineWidth", 1.5)
xlabel("Time [s]")
ylabel("Mean central \Delta Z_p")

```

```

title("Central Region Pointwise-Subtracted Vibration Signal")
grid on

w = 0.5 - 0.5*cos(2*pi*(0:N-1)/(N-1));

Y = fft(sig .* w);

P1 = abs(Y(1:floor(N/2)+1)) / sum(w);
P1(2:end-1) = 2*P1(2:end-1);

f = Fs*(0:floor(N/2))/N;

figure
plot(f, P1, "LineWidth", 1.5)
xlabel("Frequency [Hz]")
ylabel("Amplitude")
title("FFT of Central Tape Spring Region")
grid on

[~, idxMax] = max(P1(2:end));
dominantFreq = f(idxMax + 1);

fprintf("Dominant frequency = %.4f Hz\n", dominantFreq)
fprintf("Frequency resolution = %.4f Hz\n", Fs/N)
fprintf("Maximum detectable frequency = %.4f Hz\n", Fs/2)

figure
plot(nMatched, "LineWidth", 1.5)
xlabel("Frame")
ylabel("Matched ROI points")
title("Central ROI Tracking Quality")
grid on

```

Moment-Angle

```

clear; clc; close all;

%% =====
% Experimental data
% =====
experimentalFile = "bayu3point_lower_force_section_negative.csv";

S_mm = 120;
S_m = S_mm/1000;

expRaw = readmatrix(experimentalFile, "NumHeaderLines", 4);

disp_exp_mm = expRaw(:,3);
force_exp_N = expRaw(:,4);

validExp = isfinite(disp_exp_mm) & isfinite(force_exp_N);
disp_exp_mm = disp_exp_mm(validExp);
force_exp_N = force_exp_N(validExp);

theta_exp_rad = 2 * atan(2*disp_exp_mm/S_mm);
theta_exp_deg = theta_exp_rad * 180/pi;

```

```

moment_exp_Nm = force_exp_N * S_m/4;

%% Remove line after the leftmost experimental point
theta_exp_plot = theta_exp_deg;
moment_exp_plot = moment_exp_Nm;

[~, idxLeftExp] = min(theta_exp_plot);

if idxLeftExp < length(theta_exp_plot)
    theta_exp_plot = [
        theta_exp_plot(1:idxLeftExp);
        NaN;
        theta_exp_plot(idxLeftExp+1:end)
    ];

    moment_exp_plot = [
        moment_exp_plot(1:idxLeftExp);
        NaN;
        moment_exp_plot(idxLeftExp+1:end)
    ];
end

%% =====
% Abaqus data
% Column 1 = angle in rad
% Column 2 = moment
% =====
abaqusRaw = [
0      0
0.000785398  0.023832
0.0015708    0.0476469
0.00274889   0.0832943
0.00451604   0.136473
0.00716676   0.215285
0.0111428    0.3306
0.017107     0.495205
0.0260531    0.718201
0.0394724    0.977158
0.0445046    1.03722
0.0463917    1.05043
0.0492223    1.05108
0.0494877    1.04845
0.0495872    1.04702
0.0496245    1.04628
0.0496385    1.0459
0.0496464    1.04555
0.0496464    1.04555
-0.0111428   -0.262704
-0.0151189   -0.225706
-0.019095    -0.21163
-0.0230711   -0.201231
-0.0270471   -0.192021
-0.0330113   -0.179066
-0.0419574   -0.160578
-0.0553767   -0.134591
-0.0755056   -0.100476
-0.105699    -0.062133
-0.150989    -0.0330691
-0.196279    -0.0249656

```

```

-0.241569    -0.024952
-0.286859    -0.027768
-0.332149    -0.0315111
-0.377439    -0.0351736
-0.422729    -0.0380531
-0.468019    -0.0396426
-0.513309    -0.0398152
-0.558599    -0.0391262
-0.603889    -0.0384786
-0.649179    -0.0383367
-0.694469    -0.0386784
-0.739759    -0.039332
-0.785049    -0.040142
-0.785398    -0.040148
];

theta_abq_rad = abaqusRaw(:,1);
moment_abq_Nm = abaqusRaw(:,2);

theta_abq_deg = theta_abq_rad * 180/pi;

%% Remove only the connection after the positive Abaqus peak
%% Positive branch ends at the peak, then negative branch restarts from 0,0
theta_abq_plot = theta_abq_deg;
moment_abq_plot = moment_abq_Nm;

idxEndPositiveAbq = find(theta_abq_plot >= 0, 1, "last");

theta_abq_plot = [
    theta_abq_plot(1:idxEndPositiveAbq);
    NaN;
    0;
    theta_abq_plot(idxEndPositiveAbq+1:end)
];

moment_abq_plot = [
    moment_abq_plot(1:idxEndPositiveAbq);
    NaN;
    0;
    moment_abq_plot(idxEndPositiveAbq+1:end)
];

%% =====
% Analytical steady-state moments
% Signs switched to match plot convention
% =====
M_equal_analytical    = -3.37e-2;
M_opposite_analytical =  6.26e-2;

%% =====
% Combined comparison plot
% =====
figure;
hold on;
grid on;
box on;

plot(theta_exp_plot, moment_exp_plot, "LineWidth", 1.5, ...
    "DisplayName", "Experiment");

```

```

plot(theta_abq_plot, moment_abq_plot, "LineWidth", 1.5, ...
      "DisplayName", "Abaqus");

yline(M_equal_analytical, "--", ...
      "Analytical Equal-Sense", ...
      "LineWidth", 1.5, ...
      "LabelHorizontalAlignment", "left", ...
      "DisplayName", "Analytical Equal-Sense");

yline(M_opposite_analytical, "--", ...
      "Analytical Opposite-Sense", ...
      "LineWidth", 1.5, ...
      "LabelHorizontalAlignment", "left", ...
      "DisplayName", "Analytical Opposite-Sense");

xlabel("Bending Angle (deg)");
ylabel("Moment (N m)");
title("Moment-Angle Comparison");
legend("Location", "best");

```